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Topic Modeling on Noisy and Historical Swedish Documents

Evaluating Traditional and Context Based Models

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Abstract

Topic modeling is a well established technique in Natural Language Processing (NLP). Since the introduction of Latent Dirichlet Allocation (LDA), it has been a powerful tool for splitting large text collections into different categories, or “topics”, based on their content. Traditional techniques usually perform this task by modeling word frequencies in the documents. This method however, tends to struggle when handling documents riddled with noise or a very varied language. More modern approaches instead attempt to harness the abilities of contextual language models, such as BERT, to capture semantic meaning from texts. This thesis investigates how modern approaches to topic modeling compare to the traditional ones. The project uses a historical dataset, primarily in Swedish, from Riksarkivet that contains a large amount of noise from Optical Character Recognition (OCR), as well as substantial linguistic variation due to language drift over time. We evaluate four different topic models: LDA, Embedded Topic Model (ETM), Contextualized Topic Model (CTM) and BERTopic. The results show that models able to utilize semantic context perform better on the dataset at Riksarkivet compared to traditional frequency based models. BERTopic achieved the highest topic coherence, while CTM produced highly diverse topics but required substantially longer runtime. LDA remained competitive as a traditional baseline, while ETM performed worse on the evaluated metrics and was less practical in the full-dataset setting. The results also show that preprocessing remains important even for context-based models. In particular, BERTopic performed better when embeddings were generated from more heavily preprocessed data, despite the common assumption that retaining more semantic context through stopwords should benefit embedding-based methods.

Keywords

topic modeling, natural language processing, historical data, noisy data, optical character recognition, OCR, LDA, ETM, CTM, BERTopic, BERT

Sammanfattning

Studien använder ett historiskt dataset från Riksarkivet, huvudsakligen på svenska, som innehåller en stor mängd brus från optisk teckenigenkänning, samt stora språkliga variationer till följd av språkförändring över tid.

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Temamodellering är en väletablerad teknik inom språkteknologi. Sedan introduktionen av Latent Dirichlet Allocation (LDA) har metoden varit ett kraftfullt verktyg för att organisera stora textsamlingar i olika kategorier, eller ämnen, baserat på deras innehåll. Traditionellt har denna typ av modellering använt sig av ordfrekvenser i dokumenten. Detta tillvägagångssätt har visat sig ha svårigheter med att hantera texter med stora mängder brus eller har stora språkliga variationer. Nyare metoder använder istället kontextuella språkmodeller såsom BERT för att kunna använda den semantiska betydelsen i texter. Denna studie undersöker hur moderna metoder för temamodellering står sig jämfört med traditionella metoder. Studien använder ett historiskt dataset från Riksarkivet, huvudsakligen på svenska, som innehåller en stor mängd brus från optisk teckenigenkänning (OCR), samt stora språkliga variationer till följd av språkförändring över tid. Fyra olika temamodeller utvärderas: LDA, Embedded Topic Model (ETM), Contextualized Topic Model (CTM) och BERTopic. Resultaten visar att modeller som använder texternas semantiska kontext presterar bättre på Riksarkivets dataset än traditionella frekvensbaserade modeller. BERTopic uppnådde de mest sammanhängande ämnena, medan CTM producerade mycket varierade ämnen men krävde betydligt längre körtid. LDA förblev konkurrenskraftig men visade inga tydliga fördelar över de nyare modellerna, medan ETM presterade sämre enligt de utvärderade måtten och var mindre praktisk vid användning på hela datasetet. Resultaten visar även att förbehandling av data fortfarande är viktig även för kontextbaserade modeller. I synnerhet visade BERTopic förbättrade resultat när inbäddningarna genererades från mer kraftigt förbehandlad data, trots antagandet att mer bevarad semantisk kontext från t.ex. stoppord bör gynna kontextbaserade modeller.

Nyckelord

Temamodellering, Språkteknologi, Historisk data, Brusig data, OCR, optisk teckenigenkänning, LDA, ETM, CTM, BERTopic, BERT

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Acronyms and Abbreviations

BERT – Bidirectional Encoder Representations from Transformers 4, 4, 4, 4, 4, 18

BoW – Bag-of-Words 4, 4, 4, 8, 13, 15, 15, 16, 16, 16, 17, 17, 17, 18, 18, 19, 22, 27, 28, 29, 31, 31, 34, 35, 37, 40, 40, 40, 50, 58, 61

BPTM – Bayesian Probabilistic Topic Model 12

CER – Character Error Rate 24, 24, 37

D1 – Dataset 1: The lightly preprocessed dataset, retaining semantic structure while still reducing noise compared to the raw data xi, xi, xi, xi, xi, xi, xi, xi, xi, xi, xi, xi, xiii, xiii, xiii, xiii, 35, 35, 37, 38, 40, 41, 45, 46, 46, 46, 46, 47, 47, 48, 49, 50, 50, 50, 50, 50, 51, 51, 52, 52, 53, 53, 53, 53, 53, 53, 54, 54, 55, 55, 56, 56, 56, 56, 56, 56, 57, 57, 58, 77, 81, 86

D2 – Dataset 2: The heavily preprocessed dataset, removing semantic structure by removing stopwords, allowing for a Bag of Words representation

xi, xi, xi, xi, xi, xi, xi, xi, xi, xi, xi, xi, xiii, xiii, xiii, xiii, 35, 35, 35, 35, 37, 40, 40, 43, 45, 46, 46, 46, 46, 46, 47, 47, 47, 48, 48, 49, 50, 50, 50, 50, 50, 51, 51, 52, 52, 53, 53, 53, 53, 53, 53, 54, 54, 54, 55, 55, 56, 56, 56, 56, 56, 56, 56, 56, 57, 57, 58, 58, 72, 73, 77, 81, 81, 85, 86

HDBSCAN – Hierarchical Density-Based Spatial Clustering of Applications with Noise viii, xi, xi, xi, xiii, xiii, 18, 41, 41, 41, 49, 49, 49, 49, 50, 51, 51, 52, 52, 53, 63, 64, 76, 77, 84

LSI – Latent Semantic Indexing 10, 10, 10, 11

MLM – Masked Language-model 22, 22, 22, 22, 22

NLP – Natural Language Processing 4, 6, 7, 21, 22, 24, 24, 27, 28, 28

NPMI – Normalized Pointwise Mutual Information 20, 20, 42

NTM – Neural Topic Model 2, 5, 9, 9, 9, 13, 15, 18, 20, 20, 20, 20, 25, 26, 30, 31, 31, 31, 31, 32, 40, 40

OCR – Optical Character Recognition 1, 1, 3, 6, 23, 23, 23, 24, 24, 36, 37, 37

OCTIS – Optimizing and Comparing Topic Models is Simple 5, 5, 25, 42, 42, 42, 42, 42, 43, 69, 79, 79, 81, 87

pLSI – probabilistic Latent semantic indexing 11, 11, 11, 11, 12, 12

***PMI* – Pointwise Mutual Information** 20, 20, 20, 20

***SBERT* – Sentence-BERT** 7, 18, 18, 22, 23, 23, 29, 41, 41, 44

***SVD* – Singular Value Decomposition** 10, 10, 11

***TC* – Topic Coherence** xiii, xiii, xiii, xiii, xiii, xiv, xiv, xiv, 46, 50, 53, 56, 59, 61, 64, 64, 64, 74

***TD* – Topic Diversity** xiii, xiii, xiii, xiii, xiii, xiv, xiv, 46, 50, 53, 56, 59, 61, 64, 64

***TSDAE* – Transformer-based Sequential Denoising Auto-Encoder** 4, 22, 22, 40, 41

***UMAP* – Uniform Manifold Approximation and Projection** 18, 19

***VAE* – Variational Auto Encoder** 16

Chapter 1

Introduction

1.1. Background

Riksarkivet houses Sweden's largest historical archive of documents, some of which date back more than a thousand years. Its mission is not only to preserve historical records, but also to make them available to the public [4], [5].

To navigate these massive data collections, efficient information retrieval is required. Relying on simple keyword search is often insufficient. Since the information should be useful to humans, the retrieval process must be easy to understand and use. Topic modeling is a powerful tool that can help structure and organize large quantities of textual documents in an interpretable way [6]. This is achieved by assigning topics to documents based on their content. However, there are many different techniques within topic modeling, all with their own advantages and disadvantages. The dataset at Riksarkivet contains many challenging properties. Most research has focused on modern English texts [7], but the data in this case is in Swedish and contains historical variance as words change or are replaced over time. Because a large amount of the documents have been stored physically, they have been transcribed using Optical Character Recognition (OCR) [8]. This has caused the dataset to contain a large amount of noise. Tackling this dataset is a unique challenge and requires careful testing of which topic modeling techniques produce the best results.

1.2. Problem

The research question this thesis aims to answer is:

What topic modeling techniques produce the best topics on the dataset at Riksarkivet?

The technique most appropriate for the task must be able to handle noisy data caused by OCR transcription. In this context, noisy means that some words may be misspelled or sometimes complete nonsense. The techniques

must also be able to handle historical variance since many words change meaning or spelling over time.

This study investigates and compares four different modeling techniques on the specific dataset. The tested models are: Latent Dirichlet Allocation (LDA) [1], Embedded Topic Model (ETM) [2], Contextualized Topic Model (CTM) [3] and BERTopic [9]. CTM and BERTopic are modern context-based topic modeling approaches, ETM is an earlier Neural Topic Model (NTM), and LDA is an older probabilistic model used as a baseline.

The performance of these models is measured and compared primarily using two metrics: topic coherence and topic diversity. Topic coherence measures how well the words in a topic fit together, while topic diversity measures how distinct the topics are from each other. Since the purpose is to produce topics that are interpretable by humans, these metrics are used to approximate topic quality in a consistent and reproducible way. Thus, the “best” model is the model that scores the best in these metrics.

Even though LDA and ETM are older models, it is still of interest to investigate their performance. Since the dataset is not well behaved, it is important to check whether the improvements of the state-of-the-art models are reflected in the results or if the classic approaches are sufficient. CTM and BERTopic both use contextual document embeddings derived from BERT-based models and may need fine-tuning in order to better represent the documents in the dataset. This may result in models of different quality compared to other studies. For this reason, it is important to include models that lack the ability to interpret semantic meaning as comparisons.

1.2.1. Hypothesis

BERT’s ability to take context into account is expected to help mitigate the problems caused by language changing throughout history. This should make BERTopic and CTM perform better than the alternatives. While ETM is expected to outperform LDA due to its handling of rare words, the BERT-based models (CTM and BERTopic) are hypothesized to achieve the highest coherence by leveraging semantic context.

These results are expected to be shown through higher topic coherence and topic diversity scores for the better-performing models. However, the results are not expected to be overwhelmingly in favor of BERTopic and CTM, as

neural models have been shown to perform relatively worse on standard metrics despite showing better results when evaluated by humans.

1.3. Purpose

The purpose of this thesis is to provide insight into what topic modeling techniques are best suited for the dataset of Riksarkivet. By investigating how noisy data with great historical variance impacts the effectiveness of different techniques, Riksarkivet receives guidance on how to keep its infrastructure efficient for both professional and general use.

Riksarkivet is an important organization for preserving documents for future generations and making that information available digitally. As the amount of data that has to be stored at the archive has increased substantially in recent decades, the need to organize it has also increased. Riksarkivet must be able to make efficient retrieval possible for both research and general public use. Guaranteeing this service is part of its mission from the Swedish government [5].

This thesis is part of a larger project where Riksarkivet aims to improve their information retrieval capabilities in their database. Exploring the usefulness of sorting the documents into relevant topics with topic modeling is a key part of this.

Scientifically, the thesis contributes to further evaluation of the models on different kinds of datasets. Models are often evaluated on well-behaved datasets; this study aims to expand that evaluation to a more problematic one. Even though noise is a common problem in datasets, multilingual properties and especially Swedish data have less precedent in the literature. Since the data comes from a wide timeframe, this provides a somewhat novel environment for testing the performance of the different models.

1.4. Goals

The goal of this thesis is to provide, through a comparative study, an answer to what topic modeling technique is best suited for the OCR transcribed historical documents at Riksarkivet. This is achieved through the following subgoals:

1. Set up a pipeline for running the different topic models on the dataset.

2. Preprocess the data to accommodate the preferred format of the models as much as possible.
3. Evaluate the models' performance using coherence and diversity measures and compare their performance.
4. Provide clear guidance on what models are most appropriate to apply to the data at Riksarkivet or if topic modeling is a worthwhile approach at all.

1.5. Methodology

A large part of the methodology involves handling the dataset. Several of the models rely on a Bag-of-Words (BoW) representation either entirely or partially when learning or representing topics. This means that the data must be prepared both as text and as token or word-frequency representations. Additionally, the BoW representation is often impeded by stop words. Even the models that can handle stop words may benefit from their removal. Therefore, stop words are removed from the BoW representations of the dataset.

Once the dataset has been sufficiently preprocessed, a Bidirectional Encoder Representations from Transformers (BERT) model must be fine-tuned. BERT is a transformer-based language model developed by Google that produces contextual word and text representations [10]. Ideally, a BERT model pre-trained on Swedish is best. The historical Swedish data implies a domain shift from the data on which the BERT models are trained [11], [12]. Luckily, the Swedish Royal Library has trained such a model, known as KB-BERT [12]. The dataset in this thesis still represents a domain shift from what KB-BERT was trained on. The BERT model must therefore be fine-tuned before it can be utilized for both CTM and BERTopic, as is standard practice in Natural Language Processing (NLP) [13]. The fine-tuning is performed in an unsupervised manner on the same corpus as the topic modeling using Transformer-based Sequential Denoising Auto-Encoder (TSDAE), which is designed to produce document-level embeddings [14], [15].

Once the model is fine-tuned, it can be applied to the relevant models. The different models are then applied to the dataset and produce their topics. The generated topics can then be evaluated based on their most common words. This study follows standard practice and evaluates the models mainly using intrinsic measures. Although limited, they are widely used and allow

for comparisons with previous studies [6]. Specifically, topic coherence and topic diversity are the metrics used.

Topic coherence investigates the most popular words in a topic and describes how well they “fit together”. Topic diversity instead looks at how unique the topics are. High diversity means that the topics have very few words in common.

The evaluations use the Optimizing and Comparing Topic Models is Simple (OCTIS) framework. OCTIS is a framework aimed at training, analyzing, and comparing topic models and has been used in multiple previous studies [6].

1.6. Delimitations

This section describes how the scope of the thesis has been limited due to time, storage or compute restraints.

- **Evaluation:** No large-scale human evaluation is conducted. Instead, the intrinsic measures of coherence and diversity are used. This makes the evaluation simpler to conduct and allows for comparisons with previous studies, as those measures are widely used in the topic modeling domain.
- **Information retrieval:** This paper strictly focuses on the performance of topic modeling on the dataset. Even though the final goal is to later use the topics for information retrieval, this step is not investigated in this paper.
- **Models:** Due to the large number of different topic models, only a few of them are considered in this study. The selected models include one traditional probabilistic baseline and three more recent NTM or context-based approaches.

1.7. Structure of the Thesis

The thesis is divided up into the following sections:

Chapter 2 describes the relevant background needed to understand the method and model decisions. It describes all relevant topic models as well as some older ones in Section 2.1. Section 2.2 then introduces the evaluation metrics used in the thesis. It explains how the metrics work, what their strengths and weaknesses are, and what alternatives exist.

Section 2.3 and Section 2.4 go into more depth into the underlying NLP methods that work as a basis for some of the context-based models and how they are applied to them.

The concept of OCR errors is very relevant for this thesis and is explained further in Section 2.5.

Lastly, it contains multiple subsections for related work in Section 2.6. This includes studies that compare the models to each other as well as studies that evaluate the models on similarly challenging datasets.

Chapter 3 contains the main methodology. It gives a description of all the steps of the methodology in Section 3.1. It also describes the dataset in more detail and how its problematic nature is handled in terms of preprocessing in Section 3.2. This chapter contains Section 3.3, which describes the actual models and specifically how they are applied to the dataset. The same section also explains how they are set up and, in the case of contextual models, how they are fine-tuned. Section 3.4 finally goes into detail about what specific software and libraries are used. It also explains the hyperparameters and the evaluation metrics.

Chapter 4 details all the results yielded after running the experiments. It is split into two main sections. First, Section 4.1 shows the detailed results and metric values for all models and their variations individually. Then, Section 4.2 focuses on the most successful versions of all models and compares them to each other to indicate which one was the most effective at handling the specific dataset. Section 4.2.2 showcases some of the topics generated by the models

While the results section displays all the results, Chapter 5 focuses on analyzing the results and what they mean in relation to the hypothesis and research question. This chapter also attempts to view the results in a wider context to see what other conclusions can be drawn.

Chapter 6 summarizes the most important conclusions from Chapter 5 in Section 6.1 and reviews what limitations existed in Section 6.2. Finally, Section 6.3 discusses what areas could have been further investigated in this study. This is done both in the context of yielding better results and in the context of investigating the area further based on the results yielded.

Chapter 2

Background

This chapter provides insight into the field of topic modeling as well as some auxiliary fields within NLP that are relevant for some models or the data.

Section 2.1.1 gives an overview of the field of topic modeling, while Section 2.1.2 describes some of the foundational models and introduces the many shortcomings topic modeling has struggled with. Section 2.1.3 through Section 2.1.6 elaborates on all the models considered for this study by explaining how they work and how that affects their strengths and weaknesses. Section 2.2 introduces the main evaluation metrics considered in this study.

Section 2.3 and Section 2.4 introduce BERT and Sentence-BERT (SBERT), which form the neural foundation of the most modern models used in this thesis.

Section 2.5 gives an overview of what OCR-transcription is and what problems this generally causes for topic modeling.

Section 2.6 compiles multiple studies on how topic models have been used in the contexts of noisy, multilingual, and historical data. Section 2.6.1 describes how BERT, which is the neural foundation of more modern models, fares in multilingual contexts. Section 2.6.2 describes how some models have compared to each other on mostly unproblematic datasets. Section 2.6.3 reviews how previous studies handle noisy datasets for use in topic modeling. Section 2.6.4 summarizes a few studies that have applied topic modeling specifically on Swedish datasets and the problems specific to Swedish. Section 2.6.5 finally summarizes studies that have applied topic models to datasets that span larger timeframes and are historical in nature.

2.1. Topic Modeling

2.1.1. Topic Modeling Overview

In this modern age, there is a massive amount of data stored digitally. This has created a great interest in the research field of information retrieval. Within this field there is a great need for summarizing large amounts of texts in some kind of lower dimensional space. One popular method of summarizing text is topic modeling. Topic models map the textual data to lower dimensional subspaces that are represented as topics. Essentially, documents are assigned “topics” that should represent their most prominent themes based on their content. Though there are many different models for this, they all try to uncover some latent semantic structures from the text collections [16].

Many models, especially traditional ones, use the collection of words in each document to decide which topics they should be classified as. Usually a probabilistic distribution is used for this. Each topic is defined as a mixture of words, each word in the vocabulary has a certain probability for appearing in each topic. For example, the word “Bank” may have a high probability to appear in a topic called “River” and “Finance” but lower probability to appear in a topic called “Furniture”. In the same way that topics are defined as a distribution over words, the documents themselves are represented by a distribution over the topics. Every document has a certain probability to belong to every topic based on the words they contain. [7], [16].

Most models are also generative. This means that they don’t just assign a probability to the words and documents, but actually attempt to model the entire distribution so that they can potentially generate documents with similar word distributions [7].

The topics are generated in an unsupervised manner. This means that the content of the topics is learned without manual labeling. The final topics are not known beforehand, but the number of topics is in many models treated as a hyperparameter [17].

Most traditional models only consider word frequencies; this is known as the BoW model. The BoW model means that the order and overall context of the words are ignored. Observing what words are present, and how many of them there are, is often enough to achieve sufficient results [6].

Recently there has been a shift from the classic probabilistic models to the more novel NTMs. A compelling advantage with NTMs is that they don't require the same complex mathematical derivations as the probabilistic models, instead replacing it by a neural architecture. This comes with the disadvantage of less interpretability as the latent embeddings of a neural network is not easily understood. Neural models are much more flexible when it comes to the kinds of data they can handle and some are much more scalable compared to traditional models. Additionally, many NTMs can take context into account instead of just focusing on frequencies of words[6].

2.1.2. Older Models

2.1.2.1. Tf-idf scheme

Before the large surge of the probabilistic models, research focused more on the algebraic approach. They used term frequencies and classical dimensionality reduction[6].

The first implementation that paved the way for topic modeling was the tf-idf scheme from 1983 [18] . Here, term frequency (tf) and inverse document frequency (idf) are used to estimate what a document is about by highlighting the most important and unique terms. The tf value for a term “t” is calculated by counting the number of times t is used in a document “d” and dividing by the total number of different terms in the document. If $f_{x,y}$ represents the frequency of the term x in document y, it is given by the formula

$$\text{tf} = \frac{f_{t,d}}{\sum_{t' \in d} (f_{t',d})}.$$

The idf value for a term is then calculated by first counting how many documents in the corpus contain the term at least once, which yields its document frequency (df). The df value is then used to calculate the idf using the formula

$$\text{idf} = \log\left(\frac{N}{\text{df}}\right)$$

[19].

Once the final tf-idf score is obtained by multiplying the tf and idf values, we can see what terms rank the highest for every document. The overall rare

terms that are more common in the specified document rank the highest. This also allows us to store the documents more efficiently by storing the scores in a matrix where the document has a score for every term. Instead of storing all the documents with all their terms we just store a T-by-D matrix, where T contains the tf-idf value for each term and D represents all the documents. The reduction is however rather insignificant as the matrix is very sparse as most scores are zero, since most words of the vocabulary are not in every document. Additionally the results didn't yield much information about the relationships between documents. [7].

2.1.2.2. LSI

A major improvement from the tf-idf scheme was the Latent Semantic Indexing (LSI) developed in 1990. The major contribution of the LSI was to use the matrix generated by the previous method, and perform Singular Value Decomposition (SVD) on it [7]. This allows the model to not just count words, but capture broader concepts. SVD splits a matrix A into 3 matrices [20]

$$A = U\Sigma V^T.$$

U contains the “left singular vectors”, which are the eigenvectors of AA^T , a T-by-T matrix relating terms to each other. Thus, U relates the terms to the concepts. Similarly, V contains the “right singular vectors”, which are instead the eigenvectors of the D-by-D matrix $A^T A$, making V relate the documents to the concepts. Finally, Σ contains square roots of the eigenvalues of $A^T A$ and AA^T . Both matrices share the same eigenvalues since they are symmetric. These eigenvalues represent the concepts or “topics” in this case and are what allow the connection from the terms and documents to the matrix. This is the first version of something that can be described as “topics”. This also lets us disregard the eigenvalues that have a very small value, compressing the data by moving away from individual words and toward a smaller number of broader topics [21].

LSI solves the issue of synonymy in the simple tf-idf scheme. Now, two terms with the same meaning point to the same concept. It was also able to handle a degree of noise. Since only the larger eigenvalues are kept, weaker relationships generated by a few misspelled terms could be disregarded [6].

LSI however has some major drawbacks, most importantly, though it handles synonymy well, it is not able to handle homonymy (words with multiple

meanings), since every term only has a single point in the concept space. Its algebraic nature also makes the topics less interpretable to humans. This is a common problem of SVD. The fact that it requires the handling of very large matrices also makes it impractical for larger datasets.

2.1.2.3. pLSI

Some of these problems were addressed by its successor, the probabilistic Latent semantic indexing (pLSI) from 1999 [22]. The pLSI uses a statistical foundation to create a generative model, allowing it to model a more generic semantic structure [7].

Like the modern models, pLSI is unsupervised but still requires knowledge of how many topics should exist. Much like the LSI model, it still needs the term-document matrix. The pLSI handles three main variables: z for the topics, w for a term, and d for a document. It focuses on the two probabilities $P(z|d)$, the probability of each topic for every document, and $P(w|z)$, the probability of each word appearing in every topic. Based on these probabilities, Bayes' theorem can be used to calculate the probability that topic z explains term w in document d

$$P(z|w, d) = \frac{P(z|d)P(w|z)}{\sum_{z' \in Z} P(z'|d)P(w|z')}.$$

This probability is then iteratively maximized using the Expectation-Maximization (EM) algorithm [22].

What remains is the probabilities of each word appearing in each topic, and the probabilities of every topic being connected to each document. The latter is the problematic case. Since the probabilities must be stored once for every document, it scales linearly with the number of documents, making it ill-suited for large corpora. Additionally, there is no easy way to integrate a new document into the distributions without rerunning the whole algorithm [1], [7]. Despite its shortcomings, it introduced many of the core concepts still used in modern topic modeling today and solved some of the problems with previous models. The model is able to handle polysemy, meaning that words can now have different meanings. This is made possible by two different topics being able to have a high probability for the same term. Probabilistic models were the main focus of topic modeling until neural models began entering the field as well [6].

2.1.3. LDA

Latent Dirichlet Allocation (LDA), introduced in 2003 has been and still is one of the most widely used topic models [1]. LDA solidified Bayesian Probabilistic Topic Models (BPTMs) as the standard modeling approach by addressing many of the shortcomings present in previous models such as pLSI. pLSI's approach to tracking topic mixture per document " $p(z|d)$ ", was flawed as it scaled poorly and didn't allow for new documents to be integrated after training. LDA went around this issue by defining a universal distribution from which all documents are assumed to have drawn their topic mixture, namely a Dirichlet distribution [7], [16].

LDA uses a three-level hierarchical Bayesian structure with parameters on the corpus level, individual document level, and individual word level[1]. The parameters are displayed by the graphical model in Figure 2.1.

Corpus level: The outer layer defines three hyperparameters, α , β and k . α is the Dirichlet prior for the document-topic distributions. It controls the sparseness of topics in the documents; a low value leads to documents with fewer topic associations. β is the Dirichlet prior of the topic-word distributions. Similarly to α , a lower value for β corresponds to a lower amount of words associated with each topic. Lastly, the hyperparameter k is the total number of topics, also present in most other topic models.

Document level: The document level parameter is θ_d . This is the parameter which gives the whole topic distribution for a specific document d . In a corpus of 3 topics this vector might look like $\theta_d = [0, 3, 0, 6, 0, 1]$ meaning it's represented 30% by topic 1, 60% by topic 2 and 10% by topic 3. This is drawn from a Dirichlet distribution parameterized by α

$$p(\theta|\alpha) = \frac{\Gamma(\sum_{i=1}^k \alpha_i)}{\prod_{i=1}^k \Gamma(\alpha_i)} \theta_1^{\alpha_1-1} \cdot \dots \cdot \theta_k^{\alpha_k-1}.$$

Word level: The word level parameters are z_{dn} and w_{dn} . The parameter z_{dn} is the specific topic assigned to word number n in document d , sampled from θ_d . The actual word in the text is w_{dn} , which is assumed to have been sampled from β , indexed by z_{dn} .

The only parameter actually observed is w_{dn} , from which the latent variables z and θ are then inferred. This is given by the equation

$$p(\theta, z | w, \alpha, \beta) = \frac{p(\theta, z, w | \alpha, \beta)}{p(w | \alpha, \beta)}$$

where the denominator ($p(w | \alpha, \beta)$) is intractable. Though it cannot be calculated exactly it can be approximated using techniques such as variational inference or Gibbs sampling [1].

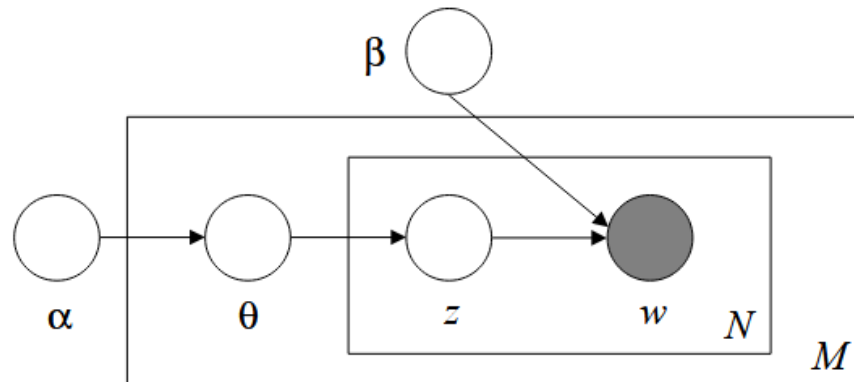


Figure 2.1: Graphical model of the LDA (from Figure 1 on page 5 of [1])

This makes LDA into a fully generative model, allowing it to apply the universal Dirichlet distribution to new, previously unseen documents. Despite this, LDA still has several drawbacks. It still heavily relies on the BoW model, preventing it from doing more than just counting the words, completely ignoring their semantic context. LDA also struggles with vocabularies containing many rare words or overrepresented words such as stop words. Additionally, even if it has the ability to handle new documents, they have to be similar to the existing documents and cannot handle a document in a new language [2], [3], [6].

2.1.4. ETM

The ETM was introduced in 2019. It was designed to handle much larger datasets than the classic LDA. It is also able to handle a corpus containing many rare words as well as stop words; some of the biggest drawbacks of the LDA. It achieves these benefits by using embeddings to represent both words and topics as coordinates in the same embedding space as seen in Figure 2.2, instead of representing each word as a discrete symbol [2]. Though it was initially described as a probabilistic model with extensions, in the context of more modern models it is now considered to be a NTM by today's standards [6], [23].

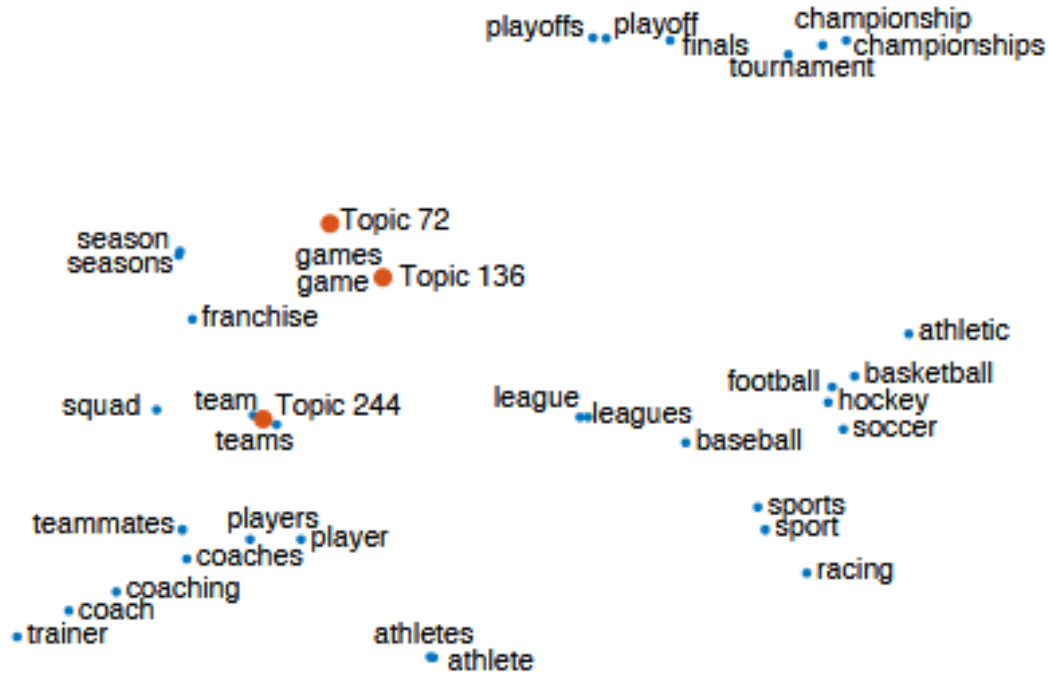


Figure 2.2: embedding space showing topics about sports (From Figure 3 on page 3 of [2])

It uses a similar generative process as the LDA in Section 2.1.3, but instead of using a Dirichlet distribution, a Logistic-Normal distribution is used. This distribution is easier to optimize later on using the reparameterization trick [2].

The ETM attempts to optimize both the word embeddings ρ and topic embeddings α to best explain the observed words in the document. This is done by maximizing the marginalized log-likelihood for all documents. The equation

$$\mathcal{L}(\alpha, \rho) = \sum_{d=1}^D \log(p(w_d | \alpha, \rho))$$

can be interpreted as “what embeddings for words and topics (α and ρ) gives the highest probability of generating the words “ w ” in all the documents “ D ”. Similarly to the LDA, this probability is intractable. The reason for this is that calculating

$$\log(p(w_d|\alpha, \rho)) = \int \left(p(\delta_d) \prod_{n=1}^{N_d} p(w_{dn}|\delta_d, \alpha, \rho) \right) d\delta_d$$

requires us to calculate every possible topic mixture for every document δ_d , which is not analytically possible.

To get around this, amortized variational inference is used [24]. The inference network is an encoder network that is optimized using the Evidence Lower Bound (ELBO) and reparameterization trick, the details of which are omitted as it is outside the scope of this paper [2].

The result of this is a coordinate vector α for every topic as well as a coordinate vector ρ for every word. Additionally, we also get the trained encoder. This encoder can then be used to classify the most prominent topics in any given document.

The encoder solution solves the previous problem of not being able to account for new unseen documents as the trained encoder can receive any document and apply its already learned embeddings [6]. It also handles stop words well as it can consider them to make up their own “junk” topic instead of confusing them with other topics [2].

This structure makes ETM theoretically suitable for large and sparse vocabularies. Despite its advantages, it still suffers from a drawback present in most contemporary models as well, the BoWs model [6], [3]. This means that it only takes into account the frequency of words in the document, not their order or context [2].

It also struggles with homonymity, when one word has multiple meanings. This is a symptom of it only having one vector embedding for every word, as it tries to push the embedding vector in two directions at the same time. This is a known problem with word2vec, which ETM uses to generate its embeddings [25].

2.1.5. CTM

The Contextualized Topic Model (CTM), introduced in 2021, is an NTM that attempts to move away from the classic BoW model [3]. Its purpose is to improve previous models by allowing them to handle semantic meaning in text to a greater extent. Specifically, the semantic meaning of a text should be considered regardless of what language it is written in. Most previous

models were monolingual [6]. The CTM showed that by partly replacing the BoW model with a neural network, a multilingual model could be achieved [3], [23].

The CTM is built with the framework of a Variational Auto Encoder (VAE). An encoder guesses the topic proportions, and a decoder reconstructs the documents based on the encoder. The framework is somewhat similar to the ETM described in section Section 2.1.4, but the input space is notably different [2]. Changing the input space is the main contribution of the CTM, as it is the major difference from neural-ProLDA, of which it can be considered an extension [3], [26].

Instead of calculating the word frequencies like in the BoW model, the document is passed through a pre-trained SBERT model to generate dense, contextualized embeddings [27]. The encoder inference network uses these embeddings as input and returns the mean and variance of a Gaussian distribution. The final topic proportions are acquired by using the reparameterization trick to add noise to the parameters and applying the softmax function, which gives

$$\theta_d = \mu_d + \sqrt{\Sigma_d} \cdot \varepsilon$$

containing the topic mixture [23], [26].

Next, the topic mixture is used as input to the decoder. In contrast to the encoder, the decoder actually uses the classic BoW model instead of SBERT. This lets the CTM learn the words in an interpretable way as opposed to the black box that is the SBERT embeddings. First, a matrix β containing the probability of every word existing within every topic is created. This is used to generate a guess of the real probabilities $\hat{x}$ using

$$\hat{x}_d = \text{softmax}(\beta\theta_d).$$

This guess is then compared against the real distribution x giving the reconstruction loss

$$\mathcal{L}_{\text{reconstruct}} = - \sum_{v=1}^V (x_{d,v} \log(\hat{x}_{dv}))$$

which is then used together with the KL-divergence to form the ELBO that is used in back propagation [23].

The overall architecture is summarized in Figure 2.3. The figure show how contextualized document embeddings are passed through the encoder to estimate the parameters of a latent topic representation, which is then sampled and used to reconstruct the document as a BoW vector. This illustrates the central idea of CTM: contextual embeddings are used for inference, while the output remains connected to an interpretable word-based representation.

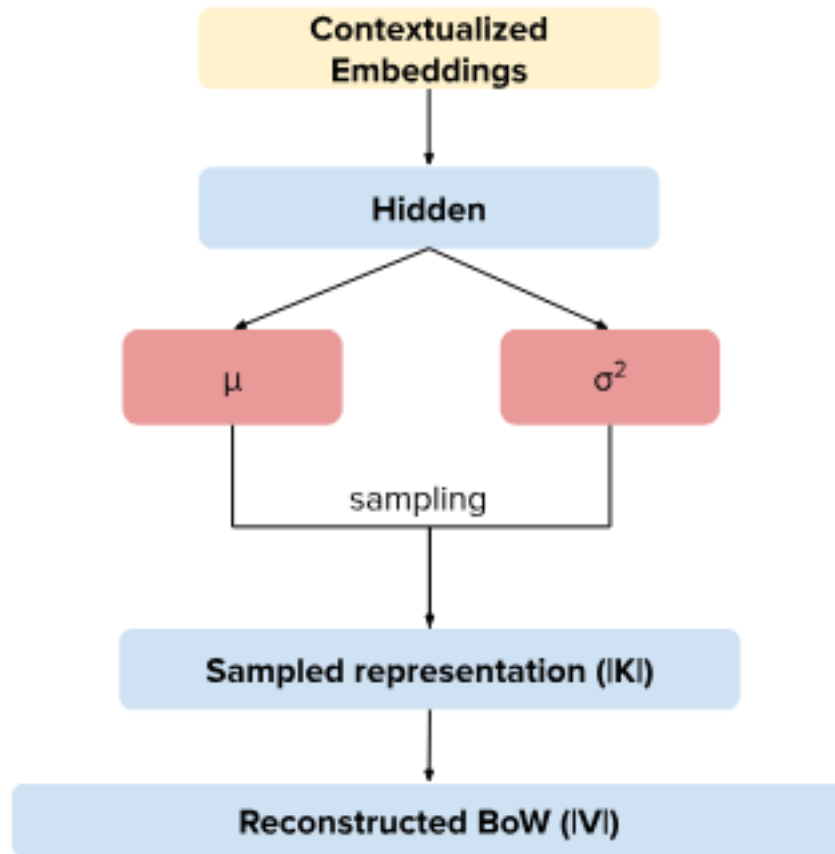


Figure 2.3: Architecture of the CTM (From figure 1 on page 2 from [3])

The training step uses both the encoder and decoder to learn the topic mixtures. After the training is done, however, only the inference using the encoder is used and outputs θ_d containing the topic mixture for the specific document. Thus, in the inference part, the BoW model is not used at all [3].

This architecture allows CTM to combine contextual document representations with interpretable topic-word distributions, since the encoder uses contextual embeddings while the decoder reconstructs a BoW representa-

tion. It also helps explain the multilingual properties of CTM, since the document representations inherit semantic information from SBERT [3].

2.1.6. BERTopic

BERTopic, introduced in 2022 attempts to tackle the problem of topic modeling from a different angle by using SBERT, an extension of BERT, described in Section 2.4 and Section 2.3 respectively, to generate embeddings. Even though the standard approach has shifted from probabilistic models to NTMs they all still share the same method of treating the topics as a probabilistic distribution which they try to reverse engineer using a generative approach. BERTopic replaces the generative approach by instead treating it as a clustering problem [9].

Similarly to the CTM, BERTopic replaces BoW with a pre-trained SBERT model in order to grasp semantic meaning and store entire documents as an embedding vector. Unlike CTM, BERTopic doesn't just partially replace the BoW model but omits it entirely for learning the topics. It only retains a word-frequency-based representation when extracting the topic words used to represent each cluster.

There are however problems when attempting clustering in high dimensional spaces. Differences in distance between close and distant pairs has been shown to shrink as dimensionality increases, making the concept of locality less defined [28]. Though there are still methods to perform clustering in these spaces, the authors of BERTopic instead opted to dimensionality reduction in order to circumvent the problem.

The method chosen for dimensionality reduction is Uniform Manifold Approximation and Projection (UMAP). It was chosen over well known methods such as PCA for its abilities to better preserve global and local features [29].

Once a low dimensional embedding for the documents is obtained, they are clustered using HDBSCAN [30]. This clustering method improves topic representations by disregarding outlier documents as noise, keeping the topic clusters coherent as they are not cluttered with loosely related documents.

After the clusters are formed, they are each assigned a topic. Instead of just naming the topics using the centroid of the clusters, BERTopic aims for more representative topic naming. BERTopic introduces a modification to the classic TF-IDF procedure introduced in Section 2.1.2.1. Instead of calcu-

lating the score for a word based on its frequency across all documents, it is calculated across the topics instead. First, all documents in a cluster are concatenated into one large document. Then, the term frequency $tf_{t,c}$ for every term “t” in the concatenated document “c” is calculated. Then, the term frequency tf_t for the same term is calculated across the whole corpus. The final score of a term is then calculated as

$$W_{t,c} = tf_{t,c} \cdot \log\left(1 + \frac{A}{tf_t}\right)$$

where A is the average number of words per cluster. The word with the highest score gives its name to the topic. In case fewer topics are desired than what was generated by UMAP, the least common topics can be merged with the most similar other topic [9].

The fact that a clustering algorithm is used means that we can skip the computational overhead that comes with the backpropagation of other models such as CTM and ETM. The topics generated are based on semantic meaning and should be language independent granted that the SBERT model used is multilingual [9], [6].

A drawback that sets it apart from other models, however, is that it assumes a single topic for every document instead of handling it as a mixture of topics. Another limitation is that even though the BoW model is unused for the actual clustering of the topics, the naming scheme for the clusters is still based on the word frequencies of the documents. This often leads to many of the top-rated words in a topic being very similar, to the point of being redundant variations of the same words [9].

2.2. Evaluation Metrics

2.2.1. Topic Coherency

Topic coherence is an evaluation metric used in topic modeling to get a grasp on how well the documents within the topics actually fit together. Previously, a measure known as perplexity was used. Perplexity measured how well the model generalizes to unseen data by testing how well the model predicts a separate set of test documents. The perplexity measure was however shown to not match with human evaluations, and topic coherence was created with the aim of bridging this gap. Coherence was meant as a measure to focus more on if the topic semantically make sense. [31], [32].

Topic coherence uses Pointwise Mutual Information (PMI) to get a statistical value of how often the N most common words in a topic actually appear together frequently in documents, compared to just independently. The PMI is calculated pairwise for two words “ w_i ” and “ w_j ” as

$$\text{PMI}(w_i, w_j) = \log \left(\frac{P(w_i, w_j)}{P(w_i) \cdot P(w_j)} \right).$$

Co-occurrence is only recognized if the words appear within a 10-word sliding window of each other.

PMI is however not sufficient as it is. The score is unbounded and shows bias towards rare words. Because of these drawbacks, modern implementation instead use Normalized Pointwise Mutual Information (NPMI) [32], [33]

$$\text{NPMI}(w_i, w_j) = \frac{\text{PMI}(w_i, w_j)}{-\log(P(w_i, w_j))}.$$

This measure normalizes the score by dividing the original PMI with the negative log of the joint probability of the same word pair. This ensures that the score is bounded between -1 and 1 . 1 indicates maximum co-occurrence, meaning that the words always appear together, 0 means that they are independent, they don’t co-occur more often than other words. -1 means that the words strictly avoid each other. The value is calculated for all $\binom{N}{2}$ pairs and then averaged [2], [31].

Topic coherence is still one of the most used metrics for topic modeling and assigns high scores to coherent and interpretable topics. Coherence has however shown some flaws when it comes to evaluating topics generated by NTMs. The NPMI assumes that the coherence of the topics must be indicated by the co-occurrence of words, which is how probabilistic topic models derive their topics, but NTMs derives them from semantic meaning. This leads to NTM sometimes scoring lower in coherence even though a human evaluator might have scored them comparatively higher [6], [34]. Similarly NTMs has been shown to sometimes score surprisingly high in coherence even though the topics appear incoherent to human evaluators. This is most likely caused by the richness of the embeddings finding connections that are not explainable to humans [34].

2.2.2. Topic Diversity

Topic diversity is a metric that aims to promote topic categorization with distinct topics. While topic coherency described in Section 2.2.1 ensures the topics are interpretable and actually share a common theme, topic diversity measures if the different topics aren't all the same [6].

For a model with K topics, let W_k^N denote the set of the top N words in topic k . Topic Diversity is then calculated as the number of unique words across all topic word lists, divided by the maximum possible number of unique words:

$$\text{TD} = \frac{|\cup_{k=1}^K W_k^N|}{K \cdot N}.$$

The denominator $K \cdot N$ represents the maximum possible number of unique words, which occurs when no top word is repeated across any topics [2]. A low score means that many top words are shared, indicating redundant topics. A high score indicates that the model has successfully generated a set of distinct topics unique from each other [2]. The lowest score is given when all topics share all their top words, and the formula shortens to $\text{TD}_{\min} = \frac{1}{K}$. The maximum score of 1 means that none of the topics share any of their top N words. Increasing the number of topics too far usually drops the score as the model forcibly divides the corpus into more topics than what is appropriate [6].

2.3. BERT

Bidirectional Encoder Representations from Transformers (BERT) is a language representation model introduced in 2018. It's meant to be used as a pretrained language model that can be used by another NLP model. Much of its claim to fame comes from its ease of use when used for fine tuning and being adapted to different domains. Its main contribution however is the fact that it is bidirectional as opposed to unidirectional. Many models before BERT were unidirectional, such as the well known Generative Pre-trained Transformer (GPT) [10].

A unidirectional model only reads the text from one direction, this means that it only learns to predict a word based on its predecessors, not what comes after. BERT aimed to remedy this shortcoming by taking both preceding and succeeding context into account. This is done by first using an encoder transformer that can perceive an entire document at once. Then it

is trained by using a Masked Language-model (MLM). The MLM randomly hides (masks) tokens in the input and sets the objective to correctly predict the missing tokens based on its context[10].

BERT also uses “next sentence prediction” where the model is trained on pairs of sentences from the corpus and is tasked with deciding which pairs are true pairs, present in the corpus and which pairs are randomly assembled [10].

This allows BERT to perceive context to a much greater extent compared to many of its contemporaries such as word2vec that struggles with polysemy [13], [25] or BoW that disregards semantics entirely [2], [35].

The pretrained model can be fine-tuned to create highly accurate and contextualized embeddings. The embeddings can be used in many NLP applications, such as grammar error detection, cross-lingual transfer learning, and tokenization, among many other tasks [13], [35]. This fine-tuning is done by either MLM or the more modern approach, TSDAE [14].

The TSDAE methods produces better embeddings for the document as a whole compared to MLM. It works by encoding a document with a large portion of masked out words into an document embedding and is then trained by having a decoder attempt to reconstruct the unmasked document. By encoding the whole document it forces a much wider context into the embeddings compared the MLM approach [14].

Despite these prominent advantages, BERT still comes with some weaknesses that limit it. BERT uses tokenization to perceive texts, but it has a hard limit of 512 tokens [13]. This limits the scope of its context perception, which affects its performance on larger documents. It requires extensive GPU resources to train the model, making it infeasible for most actors. However, since most use cases only focus on fine-tuning the model, which still may require GPU performance but substantially less, this is often not a problem [35]. Another weakness is in the domain of Semantic Text Similarity (STS), where it is very inefficient for finding the two most similar sentences in a corpus, leading to massive computational overhead [27].

2.4. Sentence-BERT

SBERT was designed in 2020 with the intent of extending the original BERT model (described in Section 2.3) to domains it was initially unsuited for.

Specifically, BERT struggles with comparing text. If BERT was tasked with finding the most similar pair of sentences in a corpus it would have to run the inference once for every possible pair. This is great for getting a deep and accurate comparison but causes a time complexity of $O(n^2)$ making the task impractical for larger scale tasks given the architecture. This extends to similar unsupervised tasks such as clustering [27].

To remedy this limitation, SBERT ensures that the sentences can be inputted individually to generate an embedding. This embedding preserves semantic meaning in a way that allows it to be compared to the other embeddings using cosine similarity [9]. This way, the comparison step is moved outside of the network. This has been shown to lower the computation time of comparing 10,000 sentences from 65 hours with BERT to 5 seconds using SBERT. This comes at a cost of slightly decreasing its capability to perform deep comparisons and might sometimes miss less obvious similarities. This is regarded as a worthwhile trade-off for leveraging the capabilities of BERT in previously unavailable domains [3], [27].

2.5. OCR

OCR is the process of scanning textual images into actual text data. This is often used so that the textual data can be further processed digitally [8]. OCR has a wide range of use cases such as transcribing text from phone cameras for instant translation, reading traffic signs and road names in autonomous vehicles and transcribing handwritten text to preserve legal or historical documents.

Traditional methods use various multi-step processes that involve separating the text from the background, isolating text from noise and formatting such as de-skewing and other morphological operations [36]. Newer methods apply deep learning approaches to simplify the process. Examples of improvements include using convolutional networks for isolating the text and auto-encoders for interpreting it [37].

Despite improvements in OCR technologies, they are still prone to including errors in their output. The quality depends greatly on the input text. Factors such as lighting, text orientation, and fonts all affect the end result [36]. The most common errors are character substitution and spacing errors. An “o” can easily be mistaken for a “0” considering variations in handwriting. Spacing between characters can also be mistakenly created or ignored,

interpreting “of the” as “ofthe” or “project” as “pro ject” [36], [38]. These errors can cause notable problems in downstream tasks depending on their scale and the context.

The quality of OCR transcribed text can be evaluated using Character Error Rate (CER). This metric measures what percentage of characters were transcribed incorrectly. CER is calculated based on the minimum required insertions “*i*”, substitutions “*s*” and deletions “*d*” to transform the original document into the OCR transcription. The formula is given by

$$CER = \frac{i + s + d}{n}$$

where n is the total number of characters [39].

Small error rates are usually manageable in the context of classical information retrieval such as keyword searches. In the context of NLP even small errors on the scale of around 5% has shown to lead to detrimental results in downstream tasks [38].

2.6. Related Works

2.6.1. Multilingual BERT

One of the main contributions BERT claims over other models for NLP is its simplistic fine-tuning. Instead of fine-tuning all of its parameters, only the last output layer is used for fine-tuning [10]. This has been utilized by many to adapt the BERT model to different domains such as biomedicine, finance, and clinical data [13]. Most notably, BERT has been fine-tuned to work in many different languages. This raises the question of how comparable the results of the fine-tuned model are to a model working in the domain it was pretrained on. This was investigated in a study from 2020 [40]. The results of the study showed that if properly pre-trained, the model consistently showed better results than if it was only fine-tuned. The sole exception to this is when training on low-resource languages. In these instances, the data is insufficient to properly pre-train the BERT model, and a fine-tune of a more general model thus provides better results [40].

This problem with general models was also noted by a Swedish study from 2023 [11] where they argued that pre-training a model on a large set of languages causes smaller languages like Swedish to be underrepresented. They proved this by training a monolingual model on only Swedish text which

consistently outperformed more general BERT models on downstream tasks [12]. They also made a point to mention that they did not only train the model on clean data. They made sure to include some messy data from OCR-transcribed documents to ensure the model got a complete view of the Swedish language [11].

2.6.2. Comparative studies

In the last decade, there has been much research into the domain of topic modeling, and many different models have been developed with different strengths and weaknesses. Thus, much effort has also been put into evaluating the models in comparison to each other. One of these studies by Abdelrazek et al. [6] divides the topic models into the domains algebraic, probabilistic, and neural. They then picked out a few models in each category and compared them to each other. In the algebraic category they used NMF [41] and LSI, for probabilistic models they used LDA and HDP, and for neural models they chose ETM and CTM. This study in particular used two mostly well-behaved English datasets to compare the models. They used *20newsgroups* and *M10*, which are commonly used in the domain as benchmark datasets. *M10* is of similar size to *20newsgroup* but contains shorter documents, with 5.91 average words per document, and more rare words, compared to 48.02 in the *20newsgroup* dataset.

They used the OCTIS framework to evaluate the models [42]. The framework provides access to many common topic models as well as automatic evaluation using many common evaluation metrics such as diversity and coherence.

The results of the study showed that in the *M10* dataset, though NTMs performed slightly better in topic coherence, algebraic and probabilistic models still showed similar results. In topic diversity the prominence of neural models was more noticeable, except for the ETM which performed very poorly. On the *20newsgroup* dataset the algebraic and probabilistic models performed better and were sometimes even outperforming the neural models.

The conclusion drawn from the study was that even though algebraic and probabilistic models sometimes perform better than neural models, neural models often outperform the other types as soon as the dataset becomes less well behaved, more sparse or just very large. This has been reflected

in the current research trends shifting to focus on neural models instead of probabilistic which previously were the most prominent in the field.

2.6.3. Studies on Noisy Datasets

Topic models are often evaluated under ideal conditions with common benchmark datasets. This does however seldom represent the real conditions under which the models are to be used. Often the datasets may be cluttered with grammatical errors or misspelling. Many studies have thus made an effort to evaluate topic models on specifically this type of data.

LDA was explicitly investigated in a study by Geeganage et al. [43] on noisy datasets and evaluated using both coherence and perplexity. The conclusion was a strong relationship between clearly defined topics and the quality of the input used to generate the topics. When the LDA was used on noisy data it often generated meaningless topics.

These problems are echoed by Denecke et al. [44] in a study that attempts to remedy these problems on a weblog dataset. They investigated the LDA but attempted to normalize the dataset by filtering out everything but the proper nouns. They argued that verbs, adjectives and adverbs often are unsuited to describe the topic of a sentence. The study proved that the LDA indeed was capable of correctly assigning topics to noisy dataset, granted that the dataset had been heavily preprocessed.

Often, texts are extracted from historical or otherwise handwritten text, which is often done using OCR as described in Section 2.5. A study by Snickars [45] used LDA on an OCR-transcribed dataset of Swedish government data containing around 3000 documents. By utilizing stop word removal and stemming to concentrate the vocabulary into only the most essential words, he was able to extract interpretable topics.

More recently the problem of noise in the dataset has been tackled by embedding-based models as a more robust solution, replacing the probabilistic models with NTMs. This topic was investigated by Egger et al. [46] in a study where they compared the results of LDA and BERTopic among other models on a dataset of Twitter posts. The results show that though all models were able to generate somewhat relevant and interpretable topics, LDA still struggled compared to BERTopic and other embedding-based models. LDA generated many broad and less interpretable topics. BERTopic, on the other hand, was able to generate more specific topics which provided useful

insights into the data. Even compared to other embedding models, BERTopic generated more diverse topics with little overlap. BERTopic also achieved this without requiring the heavy preprocessing in the form of stopword removal and lemmatization that LDA demands. The authors do, however, admit that this leads to some redundancy in the top keywords in a topic, such as “travel bubble” and “travelbubble”, and conjunction words (e.g., after, before, to, from, at), but argue that the content of a topic is still enough to determine its meaning in spite of this.

2.6.4. Topic Modeling in Swedish Data

Most research in the field of topic modeling focuses on contemporary English corpora. This leads to model performance being under-evaluated on texts in other languages. A study that attempts to investigate the performance of topic modeling on specifically Swedish data was conducted by Blad and Svensson [47]. They applied LDA and NMF on a dataset of Swedish news articles. The study, similarly to previous studies, concluded that preprocessing the data was necessary to obtain interpretable results. Despite these efforts, however, it still described the results as unsatisfactory and stated that it could only work as an inferior addition to manual categorization. The dataset being in Swedish instead of English was attributed some of the blame for the poor results. The study refers to Hedlund et al. [48], describing Swedish as different from English in the following five areas:

1. **A more complex morphology compared to English:** Some examples of this is that Swedish has multiple different plural suffixes, definite forms and inflectional forms that often require morphological analysis programs in order to normalize: If not normalized properly by lemmatization this can cause a noun to be underrepresented as its presence is spread out through its inflections.
2. **Gender features:** Swedish nouns belong to one out of two genders, en-words or ett-words. This has a beneficial impact on NLP as it may guide models as to what a sentence may be referencing back to instead of just referring to everything as “it”. However it does nothing to help BoW based models as they cannot consider this context.
3. **Derivations:** Many words in Swedish are often derived from another “base word”, for example “lära” can become “lärare”. This causes similar problems to the morphological inflections where a

single concept becomes spread out on multiple similar words, causing sparsity in the data

4. **Compound words:** Instead of using phrases of multiple nouns, Swedish is characteristic for using compound words. These words usually needs to be decomposed in preprocessing but this is not always straightforward. Some compound words modify their components, complicating the process.
5. **Homonymy and polysemy:** Polysemy, meaning two words that mean the same thing, and homonymy, meaning one word that has multiple meanings, can together be referred to as homographs. This ambiguity often leads to confusion in NLP. This phenomenon is more common in Swedish than in many other languages, with an estimated 65% of words in running texts being homographs, compared to just 50% in English. These circumstances create reason to believe that Swedish datasets benefit more from NLP models that can take context into account than English datasets, since this lets them handle the homograph problem better.

The study by Blad and Svensson [47] attempted to get around these problems by aggressive lemmatization and also filtered the dataset to only include common nouns. As mentioned before, this worked and did indeed produce sensible results. the quality did however leave much to be desired as it by itself did not provide very deep insights into the dataset that wasn't obtainable through manual means.

Another study by Bernadeta et al. [49] used topic modeling to investigate a dataset of SVT published articles about the Covid pandemic. This study used LDA, Dynamic Topic modeling (DTM) and BERTopic. To fully be able to utilize the BoW representation they attempted stemming to preprocess the data, but similarly to [47] they realized that this was not sufficient, referencing problems with word suffixes. Instead they opted to use the more advanced lemmatization tool stanza. This finally made the LDA viable for the dataset.

In order to find the optimal number of topics to generate they ran the model multiple times, using the number of topics as a hyperparameter. The study chose to use manual evaluation instead of intrinsic metrics such as coherence scores. They don't deny the usefulness of such metric but describe human evaluation as sufficient.

Though the paper mostly focused on analyzing the results of the LDA and DTM, they mention how it is worth exploring neural alternatives to the probabilistic models. They attempted to run a BERTopic model on the dataset, which yielded sensible and interpretable clustered topics, showing that BERTopic works on Swedish data. Unfortunately, they did not specify how or if they fine-tuned the underlying SBERT model. The topics were also able to be plotted over time, showing when each topic had the most articles posted.

2.6.5. Topic Modeling in Historical Data

Historical data poses multiple different challenges. Since it originates from a different timeframe, the language used is often different from modern language. Additionally, if the historical data comes from a very wide timeframe, there may also be notable language differences even within the dataset. To further complicate matters, since historical data often originates from a time before digital storage, it is not uncommon that large parts of the data have been transcribed using OCR and thus also come with many of the problems mentioned in Section 2.5.

This problem was investigated by Hall et al. [50] in a study where they tested topic models on a datasets of 19th century German and one of Ancient Greek corpora. The study used mainly LDA and found that most of the problems stemmed from the ability to properly preprocess the data to fit with the BoW representation. They concluded that even though both corpora were filled with OCR errors, they were not very prominent in the final topics. Instead the lemmatization was the bigger problem since modern software tools are not able to properly handle historical and multilingual variances. They also noted that Greek is a very context dependent language, making it a bad fit for the LDA. The results on the German dataset was however better, generating more interpretable topics according to human assessment. Although the paper points out that there was a very low degree of inter assessor agreement, questioning the validity of conclusions that can be drawn from such evaluations.

Greek showed the best result when 10-20 topics were generated, whereas German instead performed best on 70 topics. This showcases the need to vary the number of topics as a hyperparameter to find the best results.

Another study by Bodell et al. [51] analyzed a large Swedish dataset spanning 1945-2019. They focused on the issue of “language drift” where

words change meaning or spelling over time. They describe how this causes a theme to be split over multiple topics as the vocabulary changes over time. Another problem they bring up is how the topics may also change over time and how it is up to the researchers to identify the theme of the topics once they are generated. This leads to uncertainties and may also introduce elements of bias. To get around this problem while still leveraging the effectiveness of the LDA, they modified it into a semi-supervised seeded LDA. By giving the model a set of “seed words” and forcing the model to create topics with words that co-occur with those words. They used both old and new versions of the same words to force words co-occurring with both of them to belong to the same topic, instead of being fragmented across multiple topics. This showed good performance but still struggles with the problem of polysemy.

In order to achieve more useful results it would make sense to attempt to utilize the more modern NTMs. This was specifically investigated by murugaraj et al. [52] in a study where, similarly to the study by Eggar [46], they compared the classical topic modeling methods LDA and NMF to the newer embedding based methods Top2Vec and BERTopic. This study however aims to focus on specifically historic data. Their dataset consisted of newspaper articles from 1955 to 2018. They mainly used topic coherence and diversity for evaluation.

Their findings show that NMF creates more coherent topics compared to LDA, while LDA excels in diversity. Top2Vec appeared to struggle with the large quantity of data and performed worse than both NMF and LDA, showing that the flexibility of embedding-based models does not always guarantee better results. BERTopic, however, outperformed all three models, scoring higher in both coherence and diversity. Additionally, BERTopic also scaled notably better with the number of topics generated. While the classical models grew in computational cost with the increase in topics, BERTopic remained stable. Thus, the paper highly recommends BERTopic for large and heterogeneous datasets.

They managed to get around BERT’s context limit of 512 tokens by chunking larger documents into sections, then generating an embedding for each chunk before weighing them together using mean pooling. The paper also points out that though not strictly necessary, BERTopic still benefited notably from the preprocessed dataset aimed at the classical models.

2.7. Summary

To summarize, most research has shifted towards neural topic models as they more easily handle noisy data, and data of different languages. They have overall proven to be easier to use, not requiring the same amount of preprocessing as previous models which are more reliant on the BoW representation.

Despite this, many researchers using topic modeling as a tool in their research still opt to use older models like LDA or NMF. This may be because even though NTMs are more versatile, older probabilistic models still show competitive results on well behaved datasets [6]. Even on less well behaved datasets LDA have been shown to be usable by applying heavy lemmatization on the dataset [49].

This has led to the research on how well many NTMs perform to be somewhat lacking. Even if it is not a complete research gap the field would benefit from more contributions into how these models perform different circumstances.

Even if the LDA can be made to generate useful results, they still often fall short of embedding based models. The topics generated by NTMs such as BERTopic have been shown to be more distinct and more diverse, giving deeper insight into the data [6], [46]

For BoW based models such as LDA, several studies suggest that normalization and lemmatization can be as important as, or more important than, OCR noise alone [47], [50], [51]. OCR errors can often be handled with rather simple preprocessing such as stop-word removal or removing everything except common nouns. This has however proven insufficient in Swedish corpora as Swedish has much larger amount of morphological inflections and compound words than English texts [48].

The problems with not using lemmatization can be reduced by NTMs as they use embeddings to assign similar meaning to all the different versions of words. The ability to take larger contexts into account and be pretrained on similar corpora lets the models interpret meaning from the text without getting confused by misspelled or inflected nouns.

The problem of handling historic corpora add another challenge to the mix. Though lemmatization tools had previously been sufficient to help the models like LDA to handle more difficult datasets, current lemmatization

methods still struggle with handling historical and multilingual variances [50]. the performance particularly dips on more context dependent languages such as Ancient Greek.

Some have tried to handle historical contexts using manual methods such as “seeding”. This means that they provide the model with words around which the topics should be formed. This allows them to seed multiple variations of the same word into the same topic to help the model account for multiple words representing the same thing during different times in history. This does, however, require domain expertise and manual work, which makes the method less applicable in many contexts [51].

NTMs does not always perform better than classical models on historical datasets [52]. Specifically BERTopic however has been shown to perform very well on historical data, outperforming other neural models and classical models alike in both topic coherence, diversity and scalability.

When it comes to evaluation metrics, coherence is by far the most used metric, followed by diversity. Many studies however put emphasis on using human evaluation as well. As described in Section 2.2, even though coherence and diversity are the gold standard metrics in topic modeling they are not without flaws and have been shown to produce misleading results [34]. Despite this, some studies have argued that human evaluation is no better, with evaluators often giving widely different scores to the same topics [50].

Chapter 3

Method

This chapter describes the method and steps taken in the study to compare the different topic models on the dataset from Riksarkivet, as well as how these results are evaluated. Section 3.1 goes through all the steps of the method and describes the overall process. Section 3.2 goes into further detail on the data, its properties, and how it is handled, as this is a substantial part of the work in this study. Section 3.3 describes the topic models, how they were selected, and any setup needed in order to run them properly, such as fine-tuning. Section 3.4 covers the software tools and frameworks used, as well as an overview of the hyperparameters. Section 3.5 scrutinizes the robustness of the method. It covers the evaluation metrics but also the process itself and reasons about the validity and reliability of the method.

3.1. Research Process

This thesis conducted a comparative study between four different topic models: LDA, ETM, CTM, and BERTopic. The goal is to answer the research question *What topic modeling techniques produce the best results on the dataset at Riksarkivet?* as described in Section 1.2. This study uses predefined evaluation frameworks to run and evaluate the models. This means that a substantial part of the technical work focuses on how to best preprocess and apply the data to the models. The research process is described in detail to allow for reproducibility. The entire pipeline is illustrated in Figure 3.1, it follows the following overarching steps:

1. **Literature review:**

A literature study was conducted to research the relevant topics of the domain. It focused on answering the questions:

- What are the most interesting models and what are their strengths and weaknesses with regard to the dataset?
- How have other studies compared topic models before?
- How have studies in the domain of topic modeling handled similar datasets in the past?

- What evaluation metrics are common in the domain and how reliable are they?

2. **Preprocessing of data:**

The data was preprocessed in order to accommodate all the models. This requires the following steps:

- Parse the data from the OCR-transcribed XML files.
- Lightly preprocess the corpus to produce a full text version for the neural models. This includes repairing words split by line breaks and lowercasing the text
- Further preprocess the text by removing stop-words and words rendered incomprehensible by OCR noise.
- Store the data as both a version containing the full text for the BERT-based models CTM and BERTopic, and a BoW representation version for ETM and LDA.

3. **Fine-tuning of models:**

In order to ensure the BERT-based models can follow the semantic context of the dataset, the underlying BERT model must be fine-tuned to the specific dataset. A 200k sample was used for the fine-tuning

4. **Framework setup:**

Set up the evaluation framework and the pipeline to use it. This includes:

- Set up the OCTIS framework for use.
- Set up BERTopic from its original GitHub repository for use.
- Apply the preprocessed dataset to the framework.
- Apply the fine-tuned SBERT models to CTM and BERTopic.

5. **Running models:**

The models were then run; LDA, ETM, and CTM were run through the OCTIS framework, while BERTopic was run separately using its own library. The models were run for different numbers of topics, $K \in \{10, 20, 30, 40, 50, 60\}$, and their best performance was compared.

The models were run on a 150k sample of the dataset for the main experiments. All models were run on the same sample, which is independent of the 200k sample used for fine-tuning.

LDA and ETM were trained on a BoW representation created from the heavily preprocessed dataset, D2. CTM and BERTopic were evaluated using document embeddings generated from either the lightly preprocessed dataset, D1, or the heavily preprocessed dataset, D2. In both cases, topic words were extracted from tokenized text based on D2 to reduce noisy and uninformative words in the final topic representations. This made it possible to compare whether the neural models benefited more from retaining semantic context in D1 or from the stronger noise reduction in D2.

The main experiments were conducted on the 150k sample to allow repeated runs across several models and topic counts. After the main comparison, selected best-performing configurations were also applied to the full dataset to investigate scalability and full-corpus topic assignment.

6. **Evaluation:**

The results are represented by a coherence and diversity score. The scores are calculated based on the 10 most prominent words in the topic. If a topic does not contain 10 prominent words, this topic is disregarded for evaluation.

In the case of BERTopic, some documents are assigned the “junk topic” instead of being assigned a proper topic. Therefore, a separate analysis is also conducted on BERTopic to determine what proportion of the documents were assigned a non noise topic.

7. **Comparison of evaluation metrics:**

The evaluation metrics are then compared against each other to conclude which models performed the best and for what number of topics.

8. **Analysis of results:**

Finally, the results are viewed in a wider context where the models’ ease of use, data preprocessing requirements, and evaluation metrics are viewed together to draw conclusions on which models are best suited for the dataset at Riksarkivet.

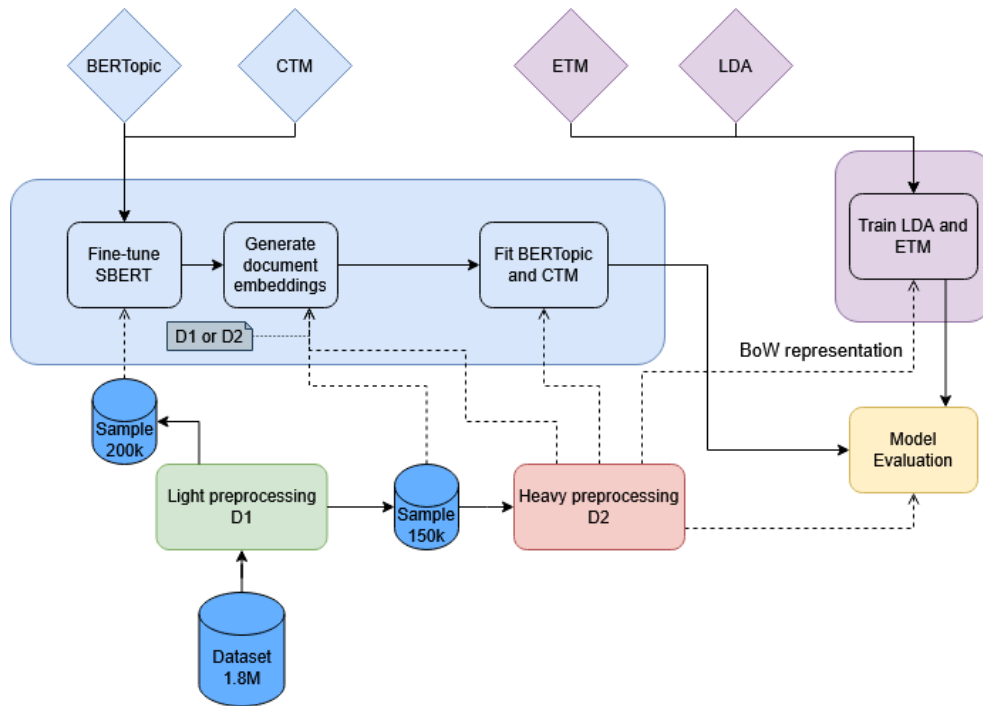


Figure 3.1: Overview of the entire project pipeline

3.2. Dataset

This section covers everything related to the dataset. Section 3.2.1 covers how the data was acquired. Section 3.2.2 covers the special properties of the dataset that the methodology needs to account for. Section 3.2.3 covers the various steps needed in order to preprocess the data.

3.2.1. Data collection

The data used in this study is provided by Riksarkivet. It is part of their public records and thus does not contain any sensitive information.

The dataset contains 1,774,919 documents, approximately 1.8 million, all of which have been OCR transcribed by Riksarkivet in the last year.

3.2.2. Data properties

This section describes the specific properties of the data that need to be taken into consideration. Section 3.2.2.1 describes how the OCR has affected the data, and Section 3.2.2.2 describes how the unique timespan of the documents affects the dataset.

3.2.2.1. OCR transcription

The data files contain text transcribed from handwritten documents using OCR. It contains most of the common errors caused by OCR described in Section 2.5, such as misspelled words or words that have been separated or mixed together. Many words are also split by line breaks. The transcribed text has an estimated CER of 0.03 to 0.1.

3.2.2.2. Historical variations

Riksarkivet stores documents as old as a thousand years. The documents in this particular dataset span roughly from the 1600s to the 1900s. This means that the Swedish used in the documents is different from modern Swedish. Since these documents span several hundred years, they also have language variation compared to each other. Additionally, they have many local variations depending on where in Sweden they originated from, since national Swedish was not standardized before the 1800s.

3.2.3. Data preprocessing

Before the data can be used by the topic models, there are several preprocessing steps that need to be taken. This section explains the preprocessing steps. As noted by previous studies [50], preprocessing steps and hyperparameters are too often omitted from studies, making comparisons and reproductions difficult.

The data is stored as XML files containing the transcriptions as well as related metadata. These files must be parsed into readable documents. The raw data is parsed into a single .jsonl file, storing one document per row. The documents are stored with their additional metadata as

```
{"doc_id": "id.xml", "source_image": "id.jpg", "processed_at":  
"date", "raw_text": "text", "word_count": 250}.
```

Because the models require different input formats, two preprocessed versions of the documents were created: a lightly preprocessed text version, D1, and a heavily preprocessed token version, D2, which was later vectorized when a BoW representation was required. The pipeline for the preprocessing is illustrated in Figure 3.2.

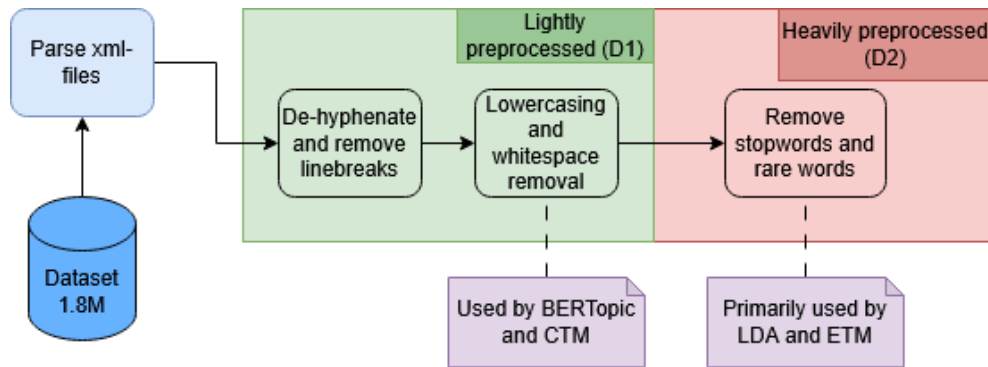


Figure 3.2: Overview of preprocessing pipeline

3.2.3.1. Dataset 1

The BERT-based models use a lightly preprocessed version of the dataset. This dataset features the following preprocessing steps:

1. **Repairing words hyphenated by line breaks:** Many words have been split up due to line breaks. An example of such a case is "närmare mid-\ndagen.", where the word "middagen" has been split into "mid" and "dagen" by "-\n". This word is stitched together using the regex: `re.sub(r'[-]\s*\n\s*', '', text)`, replacing `-\n` with the empty string. Isolated line breaks are also removed.
2. **Removing random noise:** Sometimes random symbols appear in the OCR. Therefore, every single character surrounded by white space is removed with the regex: `.sub(r'\s+[^w\s.,!?\]\s+', '', text)`. This removes some single-letter words like "ö" but was deemed to improve the overall quality.
3. **White space normalization:** All large sections of consecutive whitespace are replaced with a single white space to avoid gaps in the text.
4. **Lowercasing:** All capital letters are changed to lowercase for consistency. Other textual features, such as punctuations and space, were kept to retain the semantic structure of the text.

This yields a lightly preprocessed dataset that retains the semantic structure of the text while having a reduced amount of noise compared to the raw dataset. This dataset is referred to as D1.

3.2.3.2. Dataset 2

LDA and ETM, which require a more heavily preprocessed dataset, take the lightly processed dataset and further process it. This is done through the following steps:

1. **Noise removal:** This step removes additional OCR noise that wasn't caught in the lightly preprocessed dataset. This step focuses on removing the rarest words that have likely been created by misspellings or mistranscriptions. This is done by deleting all words that only appear in a maximum of 15 documents.
2. **Stopword removal:** Removing stopwords is done in two steps, first based on frequency, and then based on a manually compiled blacklist removing common known stopwords.
 - Since there is no comprehensive list of all historical Swedish stop words, they are removed based on frequency. This has been done by other studies [50] but with a much higher cutoff at only 50%. Since this study handles a dataset containing documents from several hundred years, the language has changed to a much higher degree. Because of this larger timeframe, a lower cutoff of around 20% is used. However, this number is treated as the hyperparameter α . This method allows for the removal of domain-specific stopwords that are overrepresented in this specific dataset.
 - To ensure a large amount of captured stopwords are removed, a list of stopwords is also compiled. Since no full list exists, the list used in this study is imperfect. First, a precomposed word list with modern Swedish stopwords is used [53]. This list is then provided to an AI agent in order to expand it with historical stop words. This added new stopwords as well as adding variations of already existing stopwords. Lastly, after running the models, additional stop words are added to the list based on manual inspection of the topics.

Unlike other studies, the data is not run through a lemmatizer even if it is recommended for Swedish corpora. This is because lemmatizers struggle with historical data and would likely cause more harm to the dataset than they would help.

This is the heavily preprocessed dataset, referred to as D2. When required by a model, D2 was converted into a BoW representation.

3.3. Topic Models

3.3.1. Model Selection

Based on previous research, neural topic models are increasingly used because they can incorporate contextual or semantic representations of text. For this reason, both CTM and BERTopic were chosen for evaluation. However, LDA has remained a relevant topic model still used in many studies. Under the right circumstances it still performs on par with, or better than NTMs. For this reason, it was important to include it in the study as a baseline. Though most recent research has focused on NTMs, they have yet to be fully adopted by researchers in other fields. For this reason, it is scientifically interesting to compare the different model paradigms to each other. ETM is a neural model that still relies heavily on the BoW representation; therefore, it is also evaluated as a middle ground between the two paradigms.

3.3.2. Model Fine-Tuning

Since LDA exclusively uses the BoW representation and lacks semantic understanding, and ETM utilizes static embeddings, they can be applied to the preprocessed dataset directly. CTM and BERTopic, however, use SBERT to grasp the semantic meaning of the text. This means that their understanding of the text is only as good as the underlying SBERT model. Though CTM has the capability for so-called zero-shot learning without fine-tuning, this relies on the pre-training of SBERT being done in the same domain, which in this case it was not [3].

To resolve this, a pre-trained BERT model for Swedish data is utilized. The model used for this is KB-BERT, which has been trained on billions of words sourced from Swedish newspapers, government reports, social media, Wikipedia, and more [12]. This model is then fine-tuned in an unsupervised manner by training it on the lightly preprocessed dataset, D1. The training is done using TSDAE. Since there is no shortage of data, a sample of 200k documents is used for the fine-tuning, but as described in [14], less than 100k is usually enough for proper domain adaptation. Increasing the amount of data leads to diminishing returns. Since we are only fine tuning and

not doing a full pretrain, one epoch was considered sufficient within the computational scope of the thesis.

The fine-tuning sample was fetched from the lightly preprocessed dataset, D1. KB-BERT was then loaded and converted into an SBERT style model by adding a mean pooling layer. This layer averages the token-level vectors produced by BERT into a single fixed-size vector for each document, allowing each document to be represented by one embedding. During TSDAE training, the documents were split into batches of 8, masked, and passed through the encoder-decoder network. After training, the final encoder was used as the SBERT model for the experiments. This model was then used to generate document-level embeddings for CTM and BERTopic.

3.3.3. BERTopic Variants

BERTopic’s HDBSCAN clustering step is known for sorting documents into a “junk” topic if they are deemed unfit for any of the “real” topics. Due to the noisy nature of the documents in the dataset, this may affect a substantial portion of the documents. To provide Riksarkivet with alternatives, two additional variations of BERTopic that mitigate this problem are evaluated.

- On top of the base BERTopic model, a version that modifies the parameters of HDBSCAN is used. Specifically, the “min_samples” variable was decreased from the standard 10 to 5. This allows points to associate with clusters despite having fewer neighbors.
- Alongside the aforementioned version, a variation is also used that keeps the base HDBSCAN parameters but forcibly reassigns all the documents to their best fitting “real” topic.

3.3.4. ETM Variants

One of the strengths of the ETM is its ability to handle larger vocabularies without loss of performance, as described by Dieng et al. [2]. Theoretically, this property should be beneficial for handling the dataset at Riksarkivet, as the abundance of words from different timeframes implies a larger vocabulary. To investigate if this property holds, two versions of the ETM are evaluated. One version uses a vocabulary of 2000, and a second version uses a much larger vocabulary of 6000.

3.4. Experimental Setup

3.4.1. Software and Frameworks

To ensure standardized comparisons, LDA, ETM, and CTM were implemented and trained using the OCTIS [42] framework, while BERTopic was run via its official library as it is not available in OCTIS. The OCTIS environment was set up via a CUDA command prompt. OCTIS uses the Gensim Python library in order to access the topic models.

To run the models quicker, KTH GPU resources were used.

3.4.2. OCTIS

OCTIS is a framework made for training, analyzing, and comparing topic models. It provides a stable pipeline to ensure reliable and reproducible results. It does not currently support BERTopic but is used to train the remaining models: LDA, ETM, and CTM.

The framework uses the Gensim library in order to access the models and their functionalities. Even though it cannot train and run the BERTopic model, it can still evaluate its results. Thus, OCTIS main role is to evaluate and compare the models.

3.4.3. Hyperparameters

There are a few hyperparameters considered in this study.

The number of topics K that the models generate. The models were tasked with generating a span of different numbers of topics, e.g. $K \in \{10, 20, 30, 40, 50, 60\}$, to see which ones yield the best results. The optimal number of topics may vary from model to model.

The cutoff percentage α for what is considered stop words in the dataset. This parameter is adjusted slightly to investigate its impact on the result.

3.4.4. Evaluation

The output of the models is evaluated using two intrinsic evaluation metrics:

- **Topic Coherence NPMI:** Measures the semantic interpretability of the topics by calculating the co-document frequency of the top 10 words in a topic. In this study, the metric is reported on a scale from -1 to 1 , where a score closer to 1 means better coherence.

- **Topic Diversity:** Measures the fraction of unique words across all generated topics to penalize models that produce redundant or collapsing topics [2]. A score closer to 0 indicates worse diversity, while a score of 1 represents the best possible result with perfectly distinct topics.

As described in section Section 2.2, these metrics are not perfect and are sometimes misleading when comparing neural models against non-neural models. The final conclusions therefore combine the aforementioned metrics with the runtime for each model and general difficulty of use. Difficulty of use may include aspects such as data preprocessing requirements and fine-tuning needed for the model to perform.

3.5. Validity and Reliability of Method

This study attempts to maximize validity by mimicking the methodologies of previous studies. None of the topic model architectures are reimplemented or modified. Instead, predefined models are used. The evaluation framework OCTIS is used to run the models. This improves the reproducibility of the study. Unfortunately, there are imperfections in the study due to the nature of the field.

The evaluation metrics coherence and diversity are imperfect and have been observed to not always correspond to better results for human interpretation. However, these metrics are among the most commonly used intrinsic evaluation metrics in the field of topic modeling. The results should therefore be viewed as a guideline and not conclusive evidence.

The study attempts to preprocess the data in a way that benefits the models as much as possible, but it is impossible to know if another kind of preprocessing would yield better results. As shown by [52], applying a neural model to a heavily preprocessed dataset without semantic context unexpectedly yielded better results. The nature of the data also makes it difficult to properly preprocess, and it is possible that methods other than the ones applied in this thesis may have been better suited for handling historical Swedish data.

To allow for fair comparisons, all models were evaluated using topic words derived from comparable token representations using D2.

The fine-tuning of SBERT models is also inherently probabilistic. Even if a reproduction is done by fine-tuning on the same dataset, there is no guarantee that the fine-tuning will be exactly the same. Despite this, the paper follows proven methods in the field, and the quality of the results should achieve a similar standard to previous studies.

Chapter 4

Results

This chapter covers all the results generated from the project. The results are presented using tables showing the raw values and bar charts for easier visual comparisons. As described in Section 3.4.4, the main evaluation metrics shown are coherence and diversity. Additionally, runtime is also shown.

Section 4.1 covers the models individually, comparing them against themselves and highlighting under which circumstances they perform best. These runs are done on a 150k sample of the dataset.

Section 4.2 summarizes the results from Section 4.1 and compares them to each other. This gives a clearer indication of which model was most suitable for the Riksarkivet dataset. First, the main comparison is shown based on the metrics in section Section 4.2.1. Then a showcase of the real generated topics is displayed in Section 4.2.2.

4.1. Individual Models

This section details the individual results and evaluated performance of all the tested models and relevant variations. The main focus is on the intrinsic metrics coherence and diversity. Additionally the recorded runtime of the models is also showcased. The main comparisons showcased in this section focus on comparing each model against its own variants and parameter settings.

For all models, this means comparing performance across different numbers of generated topics. For BERTopic and CTM, described in section Section 4.1.1 and Section 4.1.3, they are also compared based on the dataset used to generate their contextual document embeddings: the lightly preprocessed dataset described in Section 3.2.3.1 as D1, or the heavily preprocessed dataset, described in Section 3.2.3.2 as D2. BERTopic specifically is also compared against some variations of itself that use different settings to change the number of documents classified as “junk” in Section 4.1.2. The ETM, described in Section 4.1.4 is tested for two different vocabulary sizes.

Finally the LDA in Section 4.1.5 is only compared based on the number of topics.

4.1.1. BERTopic

As described in the methodology Section 3.1, BERTopic is evaluated on two different versions of the dataset. The versions are described in detail in Section 3.2. The first is the lightly preprocessed one that maximizes the semantic structures that BERTopic was designed to take advantage of: D1. The second version is the heavily preprocessed dataset, D2, which removes more noisy and less informative words at the cost of losing some semantic context.

BERTopic attempts to generate the requested number of topics, but in order for the topics to be usable for evaluation, they need at least 10 words in them. In the case of BERTopic, some topics fall short of this 10 word requirement. Thus the number of topics presented in Table 4.1 represents the target number of topics but the actual number of topics evaluated varies between runs.

Table 4.1: Table displaying mean values for metrics: Topic Coherence (TC), Topic Diversity (TD), and runtime (t) in seconds for BERTopic-Base for different number of topics (K). Results for when embeddings are based on D1 (left) and D2 (right). Highest value per column marked in bold, underscore denotes highest values across both embedding types.

D1 embeddings				D2 embeddings		
K	TC	TD	t (s)	TC	TD	t (s)
10	0.120	<u>0.973</u>	181.0	0.147	0.957	141.4
20	0.133	0.931	172.8	0.146	0.928	138.6
30	0.132	0.856	179.2	<u>0.174</u>	0.885	140.2
40	0.132	0.852	174.8	0.166	0.857	140.1
50	0.135	0.830	175.4	0.163	0.867	<u>137.6</u>
60	0.139	0.806	173.4	0.164	0.835	138.2

From Figure 4.1 we observe that it achieves its highest coherence score for $K = 30$ topics using the D2 embeddings. Most notably, the coherence score is consistently higher for the D2 embeddings compared to when using D1.

The diversity on the other hand stays more consistent across the two embeddings as well as across different K . From Figure 4.2 an inverse correlation

between diversity and K can be observed, with diversity increasing as the number of topics decrease. The difference is however not major as it only differs with 0.15 between the largest and smallest value

The runtime of BERTopic stays the same independently of K but is consistently lower for embeddings based on D2, as shown in Figure 4.3.

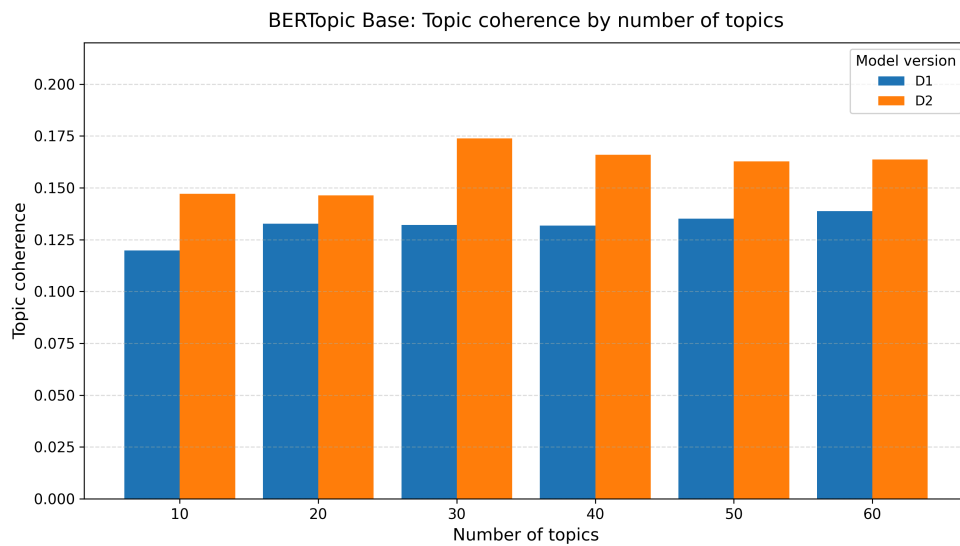


Figure 4.1: Bar chart comparing D1 and D2 embeddings based on topic coherence over different number of topics.

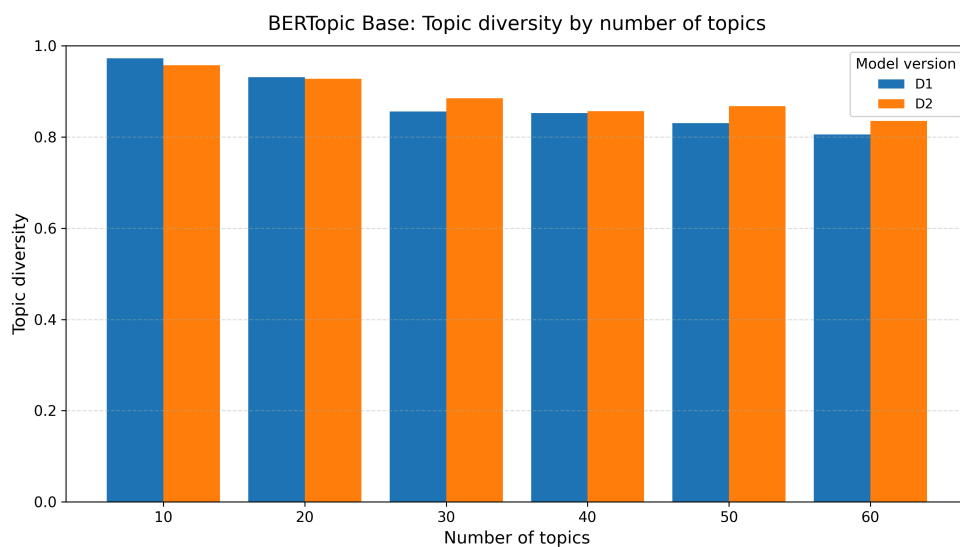


Figure 4.2: Bar chart comparing D1 and D2 embeddings based on topic diversity over different number of topics.

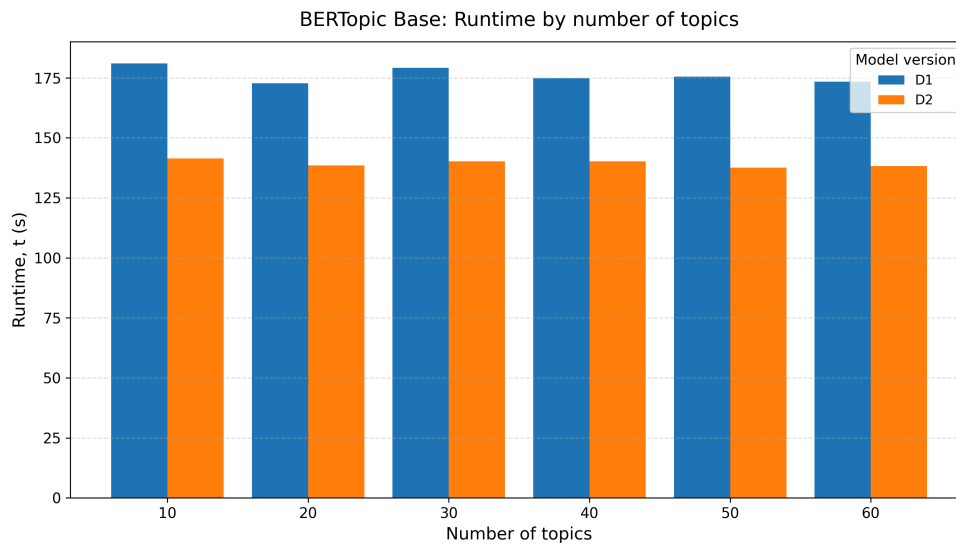


Figure 4.3: Bar chart comparing D1 and D2 embeddings based on run time over different number of topics.

BERTopic dedicates an entire topic to “junk” or “noise”. This topic is not part of the K topics listed, but is an extra topic labeled as -1 .

The results in Table 4.2 show that the number of documents classified as noise almost always is at least 50%. Overall, the D2 embeddings have a somewhat higher noise percentage but both cases are very high, except for a few outliers. These results indicate the need to investigate other BERTopic variations that decrease the high number of noise documents.

Table 4.2: Table showing percentage of noise classifications by BERTopic for both datasets. The number of topics the model was run for is given in the K column.

Embeddings	K	Total Docs	Noise Docs	Noise %
D2	60	150 000	86 743	57.8
	50	150 000	57 963	38.6
	40	150 000	86 929	58.0
	30	150 000	86 591	57.7
	20	150 000	85 137	56.8
	10	150 000	57 996	38.7
D1	60	150 000	72 252	48.2
	50	150 000	73 970	49.3
	40	150 000	71 163	47.4
	30	150 000	73 136	48.8
	20	150 000	73 954	49.3
	10	150 000	70 902	47.3

4.1.2. BERTopic variations

This section presents the results from two variations of BERTopic that aimed to reduce the amount of documents labeled as noise. The first variation, referred to as BERTopic-HDBSCAN modifies the clustering parameters of HDBSCAN to allow for more outliers to be assigned to topics. The second variation, referred to as BERTopic-Reassign simply reassigns all noise labeled documents to the closest topic after the clustering algorithm has finished.

4.1.2.1. Modified HDBSCAN parameters

The results of changing the HDBSCAN parameters are shown in Table 4.3 in the same format as in Section 4.1.1. Similarly, in many of the runs the topics did not contain enough words to meet the 10 word requirement needed for evaluation.

Table 4.3: Metrics: Topic Coherence (TC), Topic Diversity (TD), and runtime (t) in seconds for the BERTopic-HDBSCAN, for different number of topics (K). Embeddings using D1 (left) and D2 (right). Highest value per column marked in bold, underscore denotes highest values across both embedding types.

D1 embeddings				D2 embeddings		
K	TC	TD	t (s)	TC	TD	t (s)
10	0.112	<u>0.947</u>	227.9	0.125	0.918	180.7
20	0.133	0.908	205.6	0.112	0.883	<u>158.0</u>
30	0.125	0.873	194.4	0.112	0.881	160.2
40	0.124	0.857	223.5	0.111	0.877	166.4
50	<u>0.137</u>	0.842	197.4	0.080	0.858	159.1
60	0.127	0.803	198.6	0.094	0.854	179.6

The D1 embeddings now perform better than the embeddings based on the BoW representation. The coherence is lower for the D2 embeddings in all cases except for $K = 10$ as seen in Figure 4.4. Compared to the base BERTopic model, the D2 embeddings model has decreased notably, while the D1 embeddings remain somewhat similar.

In terms of diversity, Figure 4.5 shows that the values are very similar between the two embedding types. The diversity appears to slightly increase as the number of topics decreases.

In terms of runtime Figure 4.6 shows that it varies between 200 and 300 second between runs but that the model using D2 embeddings consistently runs faster than the one using the D1 embeddings. No clear correlation between time and topics is visible.

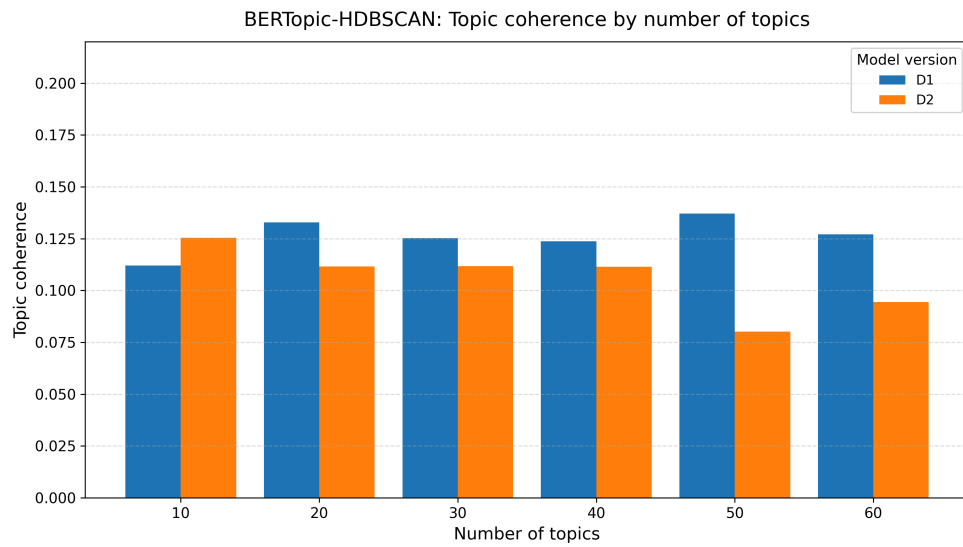


Figure 4.4: Bar chart comparing D1 embeddings and D2 embeddings for the BERTopic-HDBSCAN parameters based on coherence.

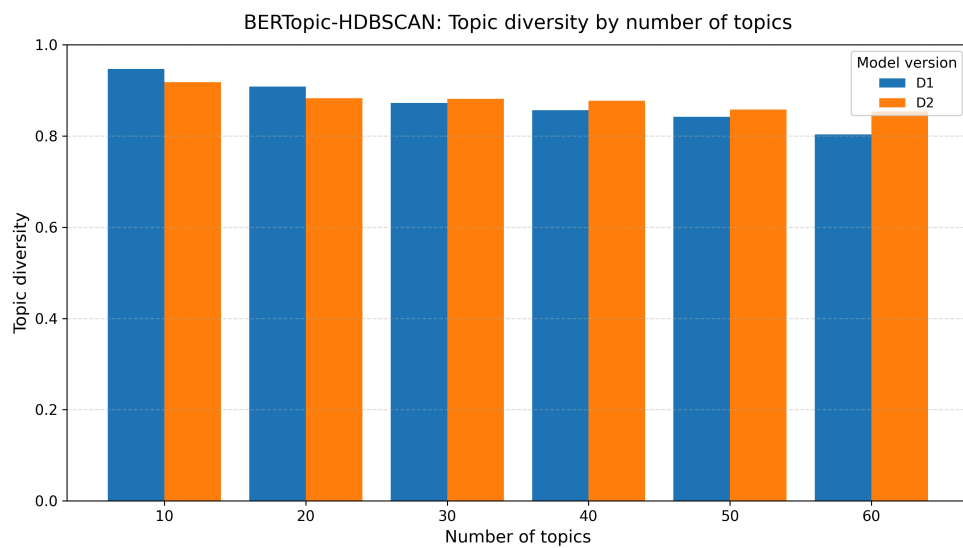


Figure 4.5: Bar chart comparing D1 embeddings and D2 embeddings for the BERTopic-HDBSCAN parameters based on diversity.

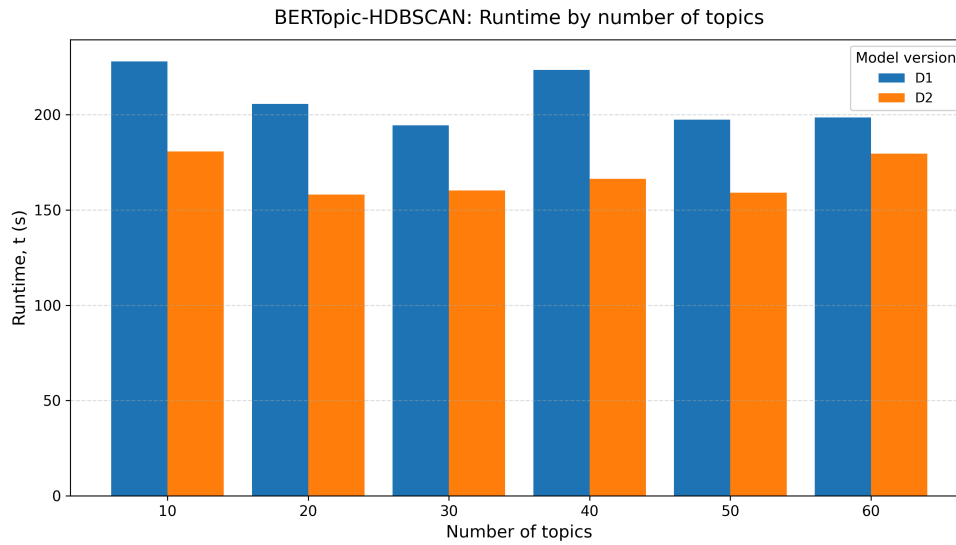


Figure 4.6: Bar chart comparing D1 embeddings and D2 embeddings for the BERTopic-HDBSCAN parameters based on run time.

Observing Table 4.4 shows that the noise in all cases stays below 40%. This is a notable decrease compared to Table 4.2, but as it still remains between 35-40%, the noise is still high compared to no noise at all.

Table 4.4: Table showing percentage of documents classified as noise by BERTopic with modified HDBSCAN parameters to reduce the amount of noise.

Dataset	K	Total Docs	Noise Docs	Noise %
D2	60	150 000	58 273	38.8
	50	150 000	58 357	38.9
	40	150 000	58 587	39.1
	30	150 000	57 602	38.4
	20	150 000	57 978	38.7
	10	150 000	58 122	38.7
D1	60	150 000	54 700	36.5
	50	150 000	56 393	37.6
	40	150 000	53 281	35.5
	30	150 000	55 612	37.1
	20	150 000	55 267	36.8
	10	150 000	55 689	37.1

4.1.2.2. BERTopic reassignment

This section shows the result of using the reassignment strategy to reduce noise. Results are presented in the same format as in Section 4.1.1. Similarly, in many of the runs the topics did not contain enough words to meet the 10 word requirement needed for evaluation.

Table 4.5 shows how the reassignment strategy retains high coherence compared to the values of the base model in Table 4.1. The diversity does however decrease somewhat. This suggests that reassignment reduced the noise rate with a smaller coherence penalty than the modified HDBSCAN setting, although it also reduced topic diversity.

Table 4.5: Metrics: Topic Coherence (TC), Topic Diversity (TD), and runtime (t) in seconds for the BERTopic-reassignment, for different number of topics (K). Results for embeddings based on D1 (left) and D2 (right). Highest value per column marked in bold, underscore denotes highest values across both embedding types

Topics	D1 embeddings			D2 embeddings		
	TC	TD	t (s)	TC	TD	t (s)
10	0.119	0.937	197.1	0.146	0.951	145.6
20	0.156	0.883	202.8	0.158	0.909	154.8
30	0.148	0.840	203.6	0.146	0.848	156.8
40	0.146	0.803	201.6	0.145	0.827	156.0
50	0.147	0.783	217.2	0.113	0.821	164.2
60	0.141	0.750	212.0	0.168	0.756	163.7

The coherence score in Figure 4.7 shows the highest values for the D2 embeddings but it is not a universal pattern; for many values of K , D1 and D2 embeddings score very similarly. For $K = 50$, D1 noticeably outperformed the D2 embeddings.

Figure 4.8 shows that the topic diversity is similar for both embedding styles. The diversity of the D2 embeddings does, however, slightly exceed the values of the D1 embeddings for all values of K . The overall diversity also trends upward as K decreases. This trend is stronger here compared to BERTopic-Base.

Figure 4.9 shows that the runtime when using the reassignment setting is similar to the base model described in Section 4.1.1. The D1 embeddings are

consistently slower than the D2 embeddings. Both embedding types appear to have increased with 10-20 seconds compared to the base model.

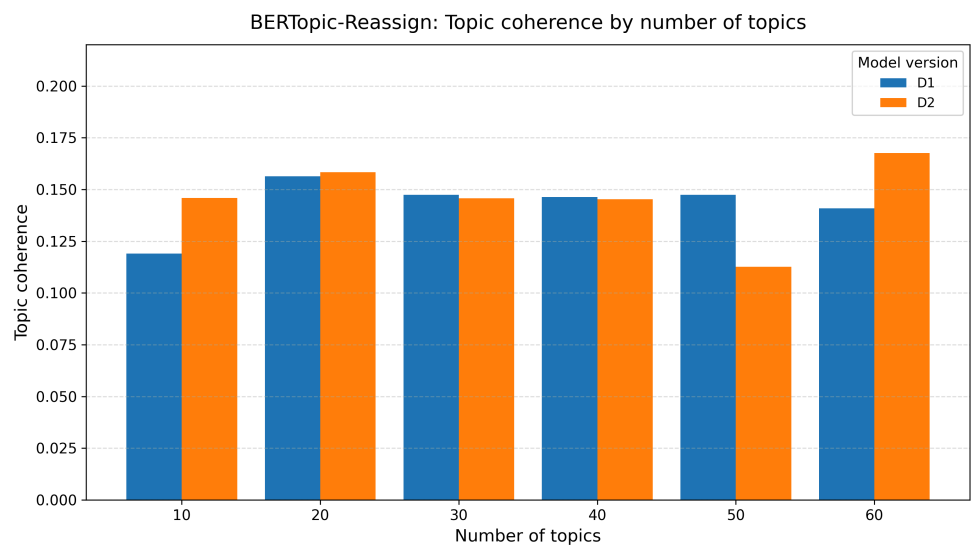


Figure 4.7: Bar chart comparing D1 and D2 embeddings for the BERTopic-Reassign based on coherence for different number of topics.

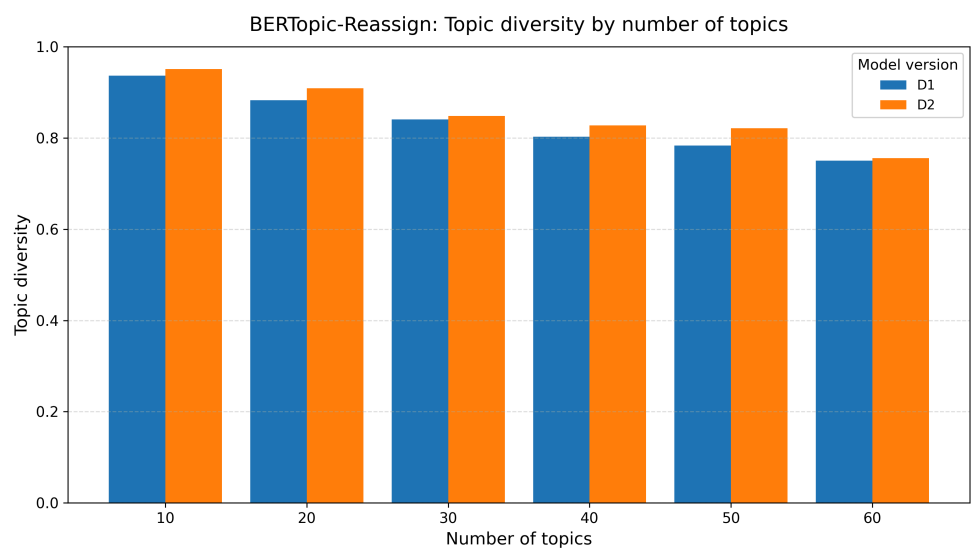


Figure 4.8: Bar chart comparing D1 and D2 embeddings for the BERTopic-Reassign based on diversity for different number of topics.

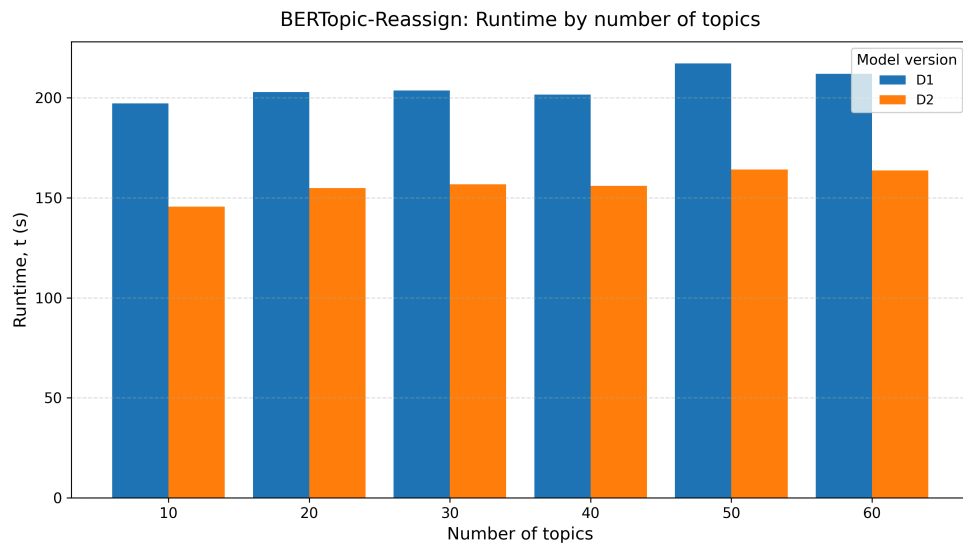


Figure 4.9: Bar chart comparing D1 and D2 embeddings for the BERTopic-Reassign based on runtime for different number of topics.

The amount of documents classified as noise have with the reassignment setting been reduced to 0% as seen in table Table 4.6. This is expected as it forcefully reassigns the documents to their nearest cluster.

Table 4.6: Table showing percentage of documents classified as noise by BERTopic-Reassign.

Embeddings	K	Total Docs	Noise Docs	Noise %
D2	60	150 000	0	0.0
	50	150 000	0	0.0
	40	150 000	0	0.0
	30	150 000	0	0.0
	20	150 000	0	0.0
	10	150 000	0	0.0
D1	40	150 000	0	0.0
	30	150 000	0	0.0
	20	150 000	0	0.0
	60	150 000	0	0.0
	50	150 000	0	0.0
	10	150 000	0	0.0

4.1.3. CTM

The CTM, much like BERTopic, uses contextual document embeddings to create its topics. Thus it also compares the two embedding strategies of utilizing D1 or D2 in addition to the comparison over the number of topics K , as seen in Table 4.7.

Table 4.7: Metrics: Topic Coherence (TC), Topic Diversity (TD), and runtime (t) in seconds for the CTM model for different number of topics (K). Embeddings based on D1 (left) and D2 (right). Highest value per column marked in bold, underscore denotes highest values across both embedding types.

K	D1 embeddings			D2 embeddings		
	TC	TD	t (s)	TC	TD	t (s)
10	0.109	<u>0.993</u>	1533.6	0.111	0.990	1513.3
20	<u>0.139</u>	0.960	1480.1	0.138	0.970	1497.3
30	0.134	0.932	1513.8	0.129	0.931	1470.1
40	0.126	0.902	<u>1464.2</u>	0.127	0.911	1488.4
50	0.114	0.892	1528.9	0.118	0.891	1502.7
60	0.113	0.864	1483.7	0.115	0.862	1490.6

Observing the CTM results in Table 4.7 and Figure 4.10, we can conclude that the CTM achieves its highest coherence for around 20-40 topics. For both the version using the D1 and D2 embeddings, the highest coherence occurred at 20 topics. The difference between using D1 and D2 for embeddings is very small as they score very similarly across all topics.

The topic diversity as shown in Figure 4.11, seems to consistently be slightly higher when using D2. But both embeddings do achieve similarly high values. Visible in both datasets is also a trend of increasing diversity as the number of topics decreases.

The runtime for the CTM is much higher than the other models. This is caused by the fact that the CTM is a neural network and the network has to be trained to fit the data every run, which takes a substantial amount of time. Runtime varied only slightly across different values of K , with no clear relationship between runtime and number of topics as seen in Figure 4.12.

The runtime remained very similar between both the D1 and D2 dataset, but was slightly lower when utilizing the D2 dataset for the embeddings when inspecting the values in Table 4.7.

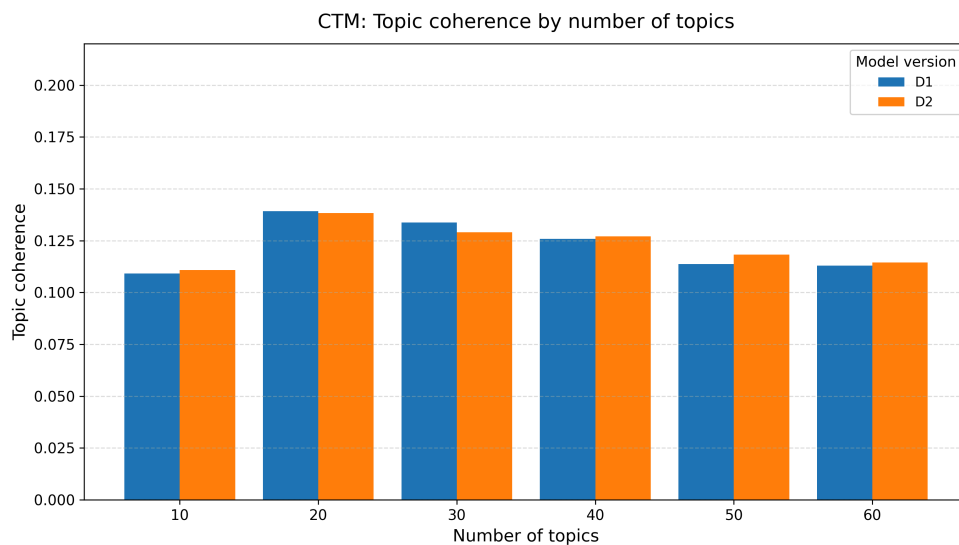


Figure 4.10: Bar chart showing a comparison of using the D1 embeddings and D2 embeddings for CTM based on coherence over different number of topics.

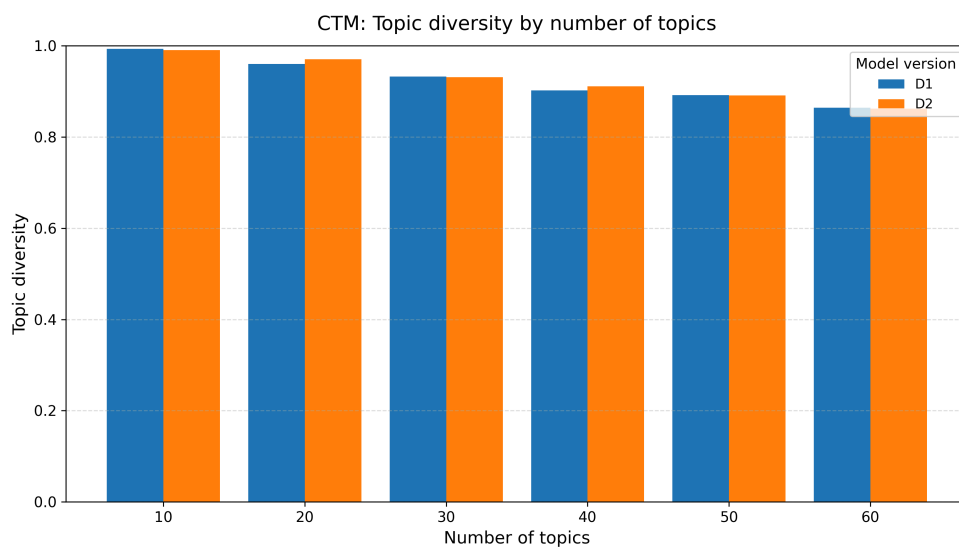


Figure 4.11: Bar chart showing a comparison of using the D1 embeddings and D2 embeddings for CTM based on diversity over different number of topics.

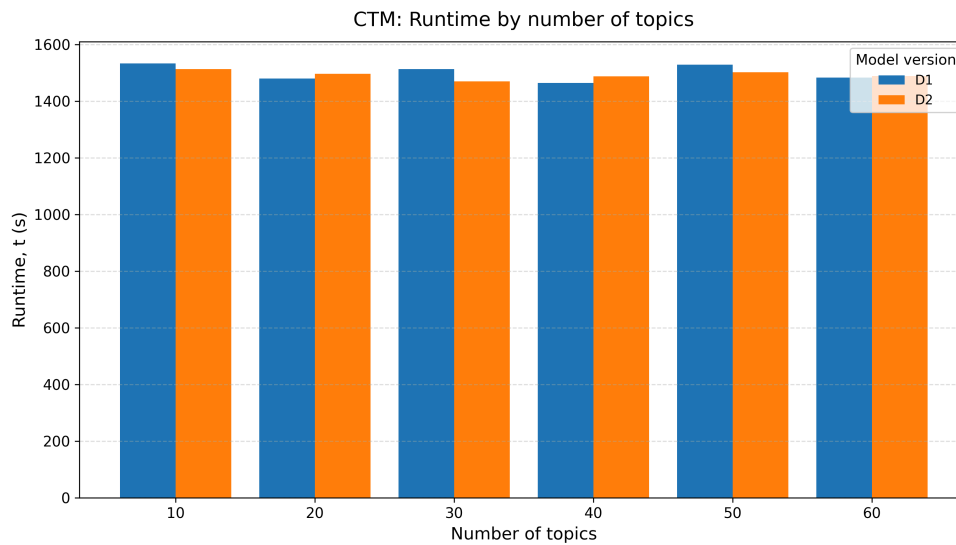


Figure 4.12: Bar chart showing a comparison of using the D1 embeddings and D2 embeddings for CTM based on run time over different number of topics.

4.1.4. ETM

ETM was evaluated using D2. Unlike CTM and BERTopic, ETM does not use contextual document embeddings, but instead learns topic and word embeddings from the BoW representation during training. Thus, unlike CTM and BERTopic, ETM learns these embeddings during each run rather than using precomputed document embeddings.

ETM has a fixed vocabulary size. One of its strengths compared to LDA is that it should be able to handle a larger vocabulary. For this reason, the results in Table 4.8 are compared based on the number of topics generated as well as two different vocabulary sizes, 2000 words and 6000 words.

Table 4.8: Metrics: TC, TD, and runtime (t) in seconds for the ETM for different K . Results for vocabulary size 2000 (left) and 6000 (right). Highest value per column marked in bold, underscore denotes highest values across both embedding types.

Topics	V2000			V6000		
	TC	TD	t (s)	TC	TD	t (s)
10	0.080	<u>0.900</u>	430.1	0.063	0.853	620.0
20	<u>0.083</u>	0.685	432.9	0.073	0.722	571.1
30	0.071	0.533	438.1	0.058	0.546	556.7
40	0.069	0.441	430.5	0.054	0.452	<u>544.3</u>
50	0.067	0.367	429.8	0.061	0.427	616.3
60	0.064	0.312	424.1	0.054	0.359	615.1

Observing the coherence score displayed in Figure 4.13, the ETM appears to score about the same across most topics, but slightly higher at $K = 20$. Utilizing a 2000 word vocabulary scores higher than the 6000 word vocabulary for all numbers of topics. The highest score of 0.083 was achieved for $K = 20$ topics.

The diversity is shown as a bar chart in Figure 4.14 and for both vocabulary sizes it shows a clear decrease as the number of topics increases. The 6000 word vocabulary scores higher in diversity in all cases except for $K = 10$ topics, where it is surpassed by the 2000 word vocabulary. 10 topics is also where ETM has its overall highest diversity.

The runtime visualized in Figure 4.15 differs between the two vocabulary sizes. The smaller vocabulary stays at 420-440 seconds runtime for all topics while the larger vocabulary causes the model to sometimes take over 600 seconds to finish. The smaller vocabulary was also more stable in this regard as the larger vocabulary varied between 550-650 depending on the number of topics generated. the 2000 word vocabulary version was indeed much more stable but by observing the values in Table 4.8 they increased with the number of topics.

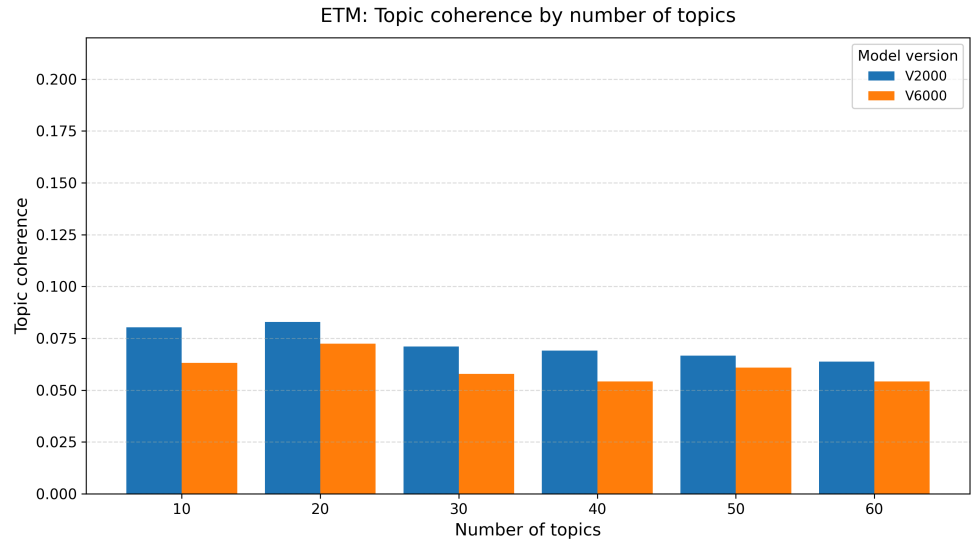


Figure 4.13: Bar chart showing the topic coherence of the ETM for 6 different number of topics and 2 vocabulary sizes

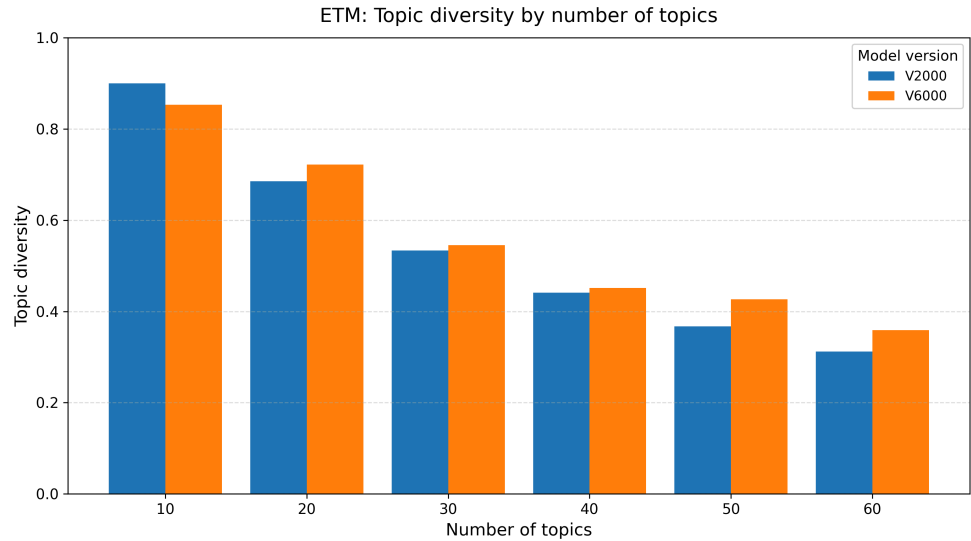


Figure 4.14: Bar chart showing the topic diversity of the ETM for 6 different number of topics and 2 vocabulary sizes

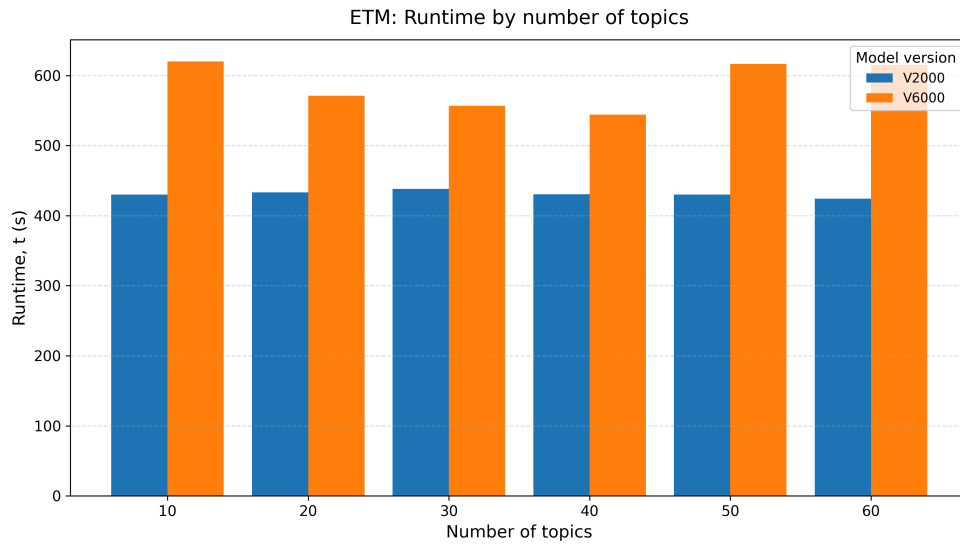


Figure 4.15: Bar chart showing the run time of the ETM for 6 different number of topics and 2 vocabulary sizes

4.1.5. LDA

LDA uses pure word frequencies instead of embeddings, thus it only uses the BoW dataset. For this reason LDA is only compared against itself based on the number of topics generated in Table 4.9.

Table 4.9: Metrics: Topic Coherence (TC), Topic Diversity (TD), and runtime (t) in seconds for the LDA model, for different number of topics (K). Highest value per column is underscored and marked in bold.

BoW preprocessing			
Topics	TC	TD	t (s)
10	0.083	0.880	<u>234.1</u>
20	0.108	0.900	234.4
30	<u>0.135</u>	0.887	242.4
40	0.117	0.875	448.0
50	0.127	0.896	299.5
60	0.111	<u>0.913</u>	297.2

The LDA had its highest coherence score of 0.135 at 30 topics. The score was varied across different K . As seen in Figure 4.16 it appeared to score higher between 30-50 topics and scored particularly low at $K = 10$.

In terms of diversity the score was very consistent around 0.9 across all K as seen in Figure 4.17. The highest score at 60 topics only differed from the lowest at 10 topics with about 0.033.

Figure 4.18 shows how the runtime of the LDA increased with the number of topics. The runtime was unusually high at 40 topics.

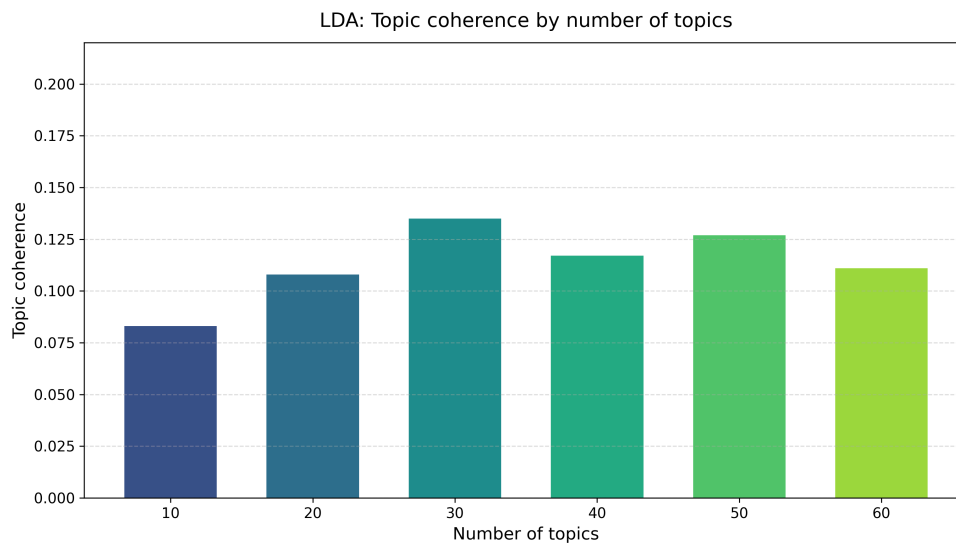


Figure 4.16: Bar chart showing the topic coherence of the LDA for 6 different number of topics generated

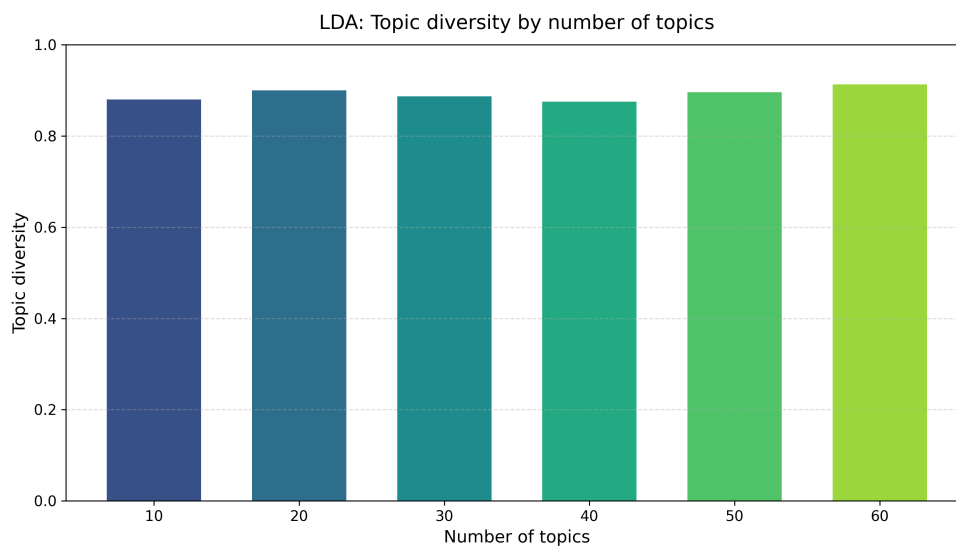


Figure 4.17: Bar chart showing the topic diversity of the LDA for 6 different number of topics generated

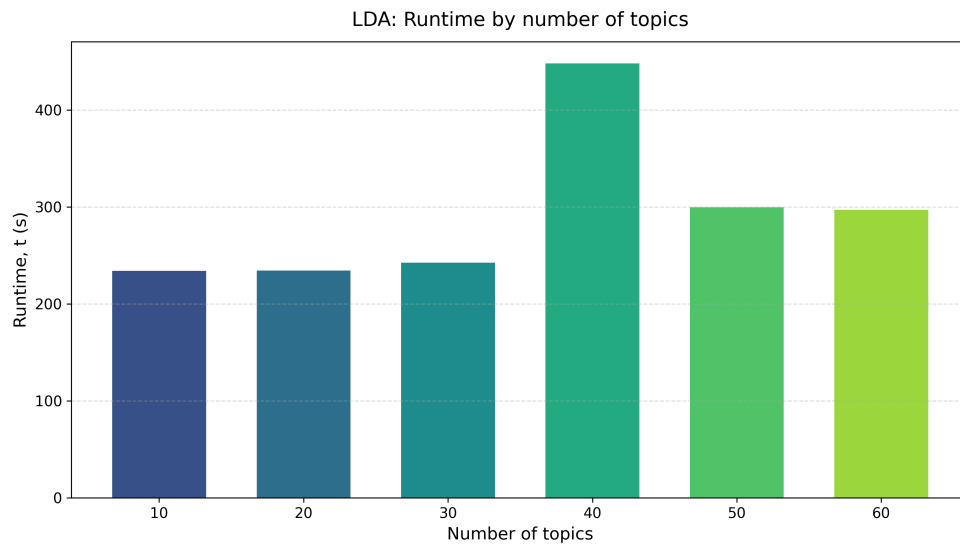


Figure 4.18: Bar chart showing the runtime of the LDA for 6 different number of topics generated

4.2. Summary of model results

This section aims to showcase how the different models compare to each other. These results guide the answer to the research question described in Section 1.2. Since the models performed differently under different circumstances, this section focuses on comparing them at their best. This means that if a model performed well for a certain number of topics but poorly for another, the better result is considered.

4.2.1. Metric Comparison

Topic coherence is treated as the primary selection metric in this section, since it is intended to capture the interpretability of topics, as described in Section 2.2. The models and their specific configuration is listed in Table 4.10

All models only contribute with one version of themselves except for BERTopic. BERTopic has two additional variations, BERTopic-HDBSCAN presented in Section 4.1.2.1 and BERTopic-Reassign presented in Section 4.1.2.2. This is because their usefulness for reducing noise must be weighted against their performance on the other metrics.

Table 4.10: Best result selected for each model according to highest Topic Coherence (TC). The table shows the selected model configuration, number of topics (K), Topic Coherence (TC), Topic Diversity (TD), and runtime (t) in seconds.

Model	K	TC	TD	t (s)
BERTopic Base (D2)	30	0.174	0.885	140.2
BERTopic-Reassign (D2)	60	0.168	0.756	163.7
CTM (D1)	20	0.139	0.960	1480.1
BERTopic-HDBSCAN (D1)	50	0.137	0.842	197.4
LDA	30	0.135	0.887	242.4
ETM (V2000)	20	0.083	0.685	432.9

The coherence score of the models is plotted in Figure 4.19. The bar chart shows the base BERTopic in the lead, followed by BERTopic-Reassign. After a drop in coherence, CTM is the third best performing. Changing the HDBSCAN parameters of BERTopic made it drop down to fourth place. The fifth best performing model was the LDA, which still scored very similarly to CTM. After another visible drop in score, the ETM had the lowest coherence among the selected models.

It is also worth noting that most models with the exception of the BERTopic variants scored the best when generating a smaller number of topics, $K \leq 30$.

The same models as presented in Figure 4.19 but instead with their diversity displayed is shown as a bar chart in Figure 4.20. The chart shows that the CTM scored the best in diversity followed by the LDA and the base BERTopic model. These were then followed by the two BERTopic variants. ETM scored the worst in diversity.

A summary of the metrics can be seen in Figure 4.21, where diversity is plotted against coherence so that the tradeoff can be seen.

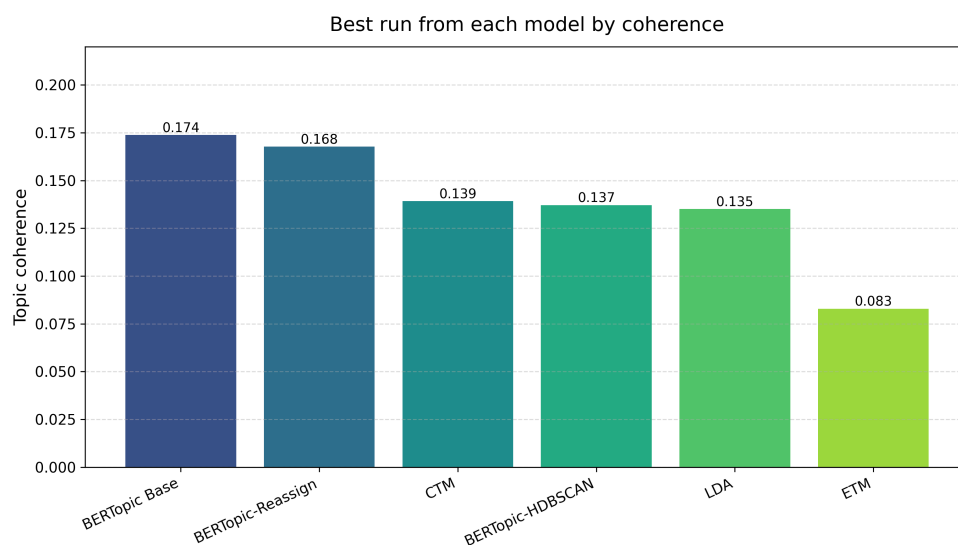


Figure 4.19: The best score in terms of coherence for all models evaluated. Presented with a bar chart

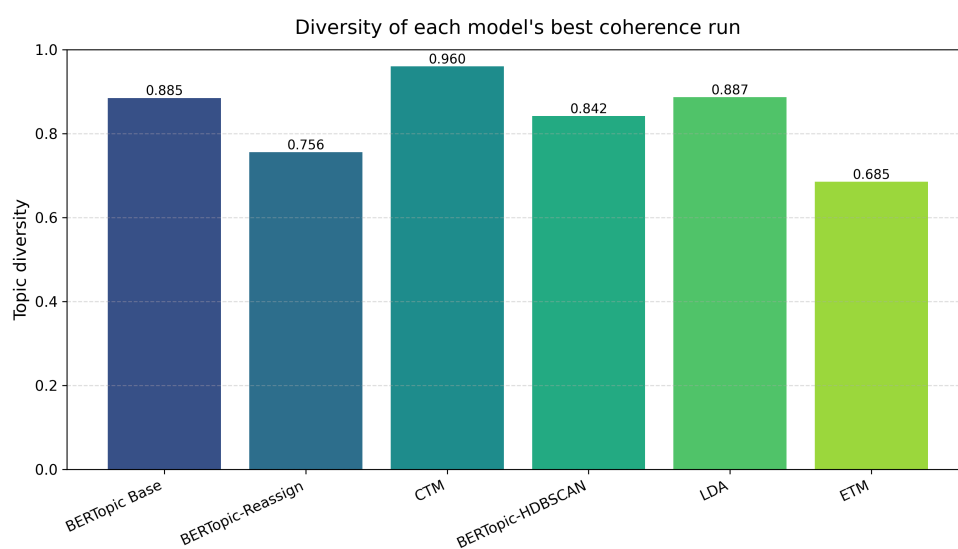


Figure 4.20: The topic diversity of the best models based on coherence. Presented as a bar chart.

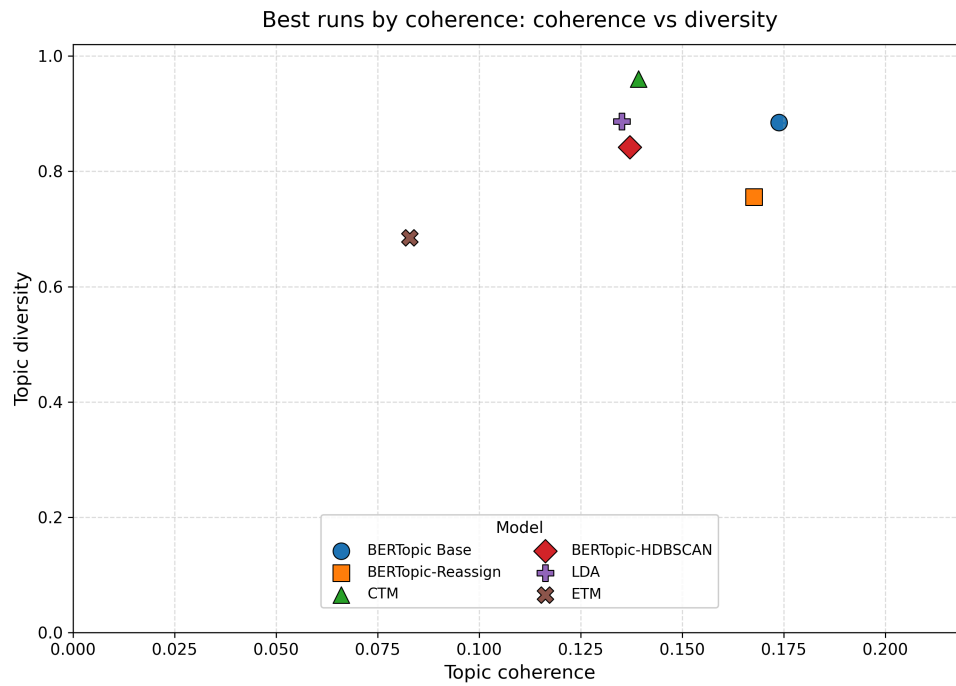


Figure 4.21: Plot showing diversity against coherence for the models.

A comparison of the models based on runtime is shown in Figure 4.22. The base BERTopic model had the shortest runtime followed by its two variants. LDA took slightly longer, followed by the ETM which took over 2 extra minutes over the LDA. Lastly the CTM had the longest runtime by a large margin, reaching approximately 25 minutes.

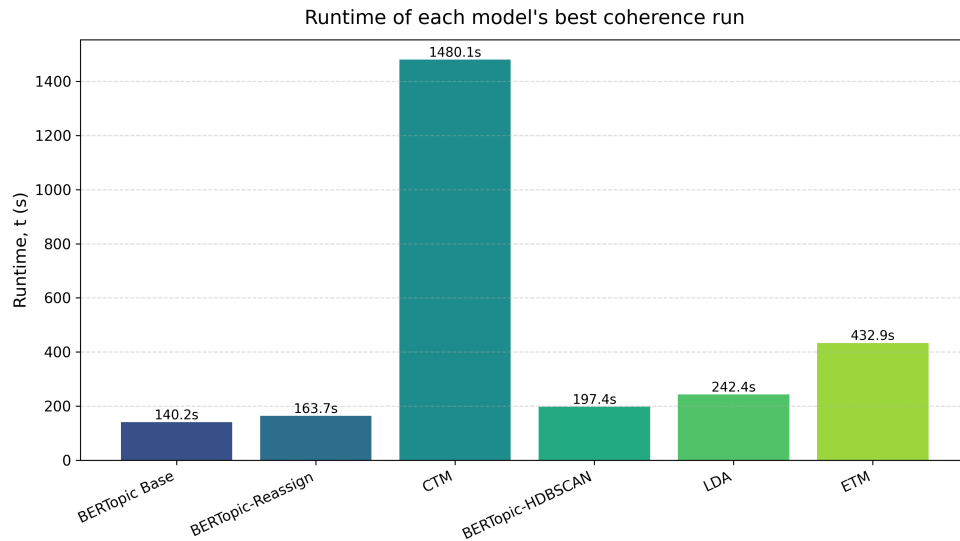


Figure 4.22: The models with the best coherence score shown by runtime in a bar chart

4.2.2. Topic Showcase

This section gives a qualitative overview of what the generated topics look like by showcasing some examples of both coherent and weaker topics in Table 4.11. The topics are taken from the best performing model runs with the configurations shown in Table 4.10. For BERTopic, the base model is used.

From the table, topics with clear semantic themes can be identified, such as legal proceedings, financial claims, household items, mining, and military. At the same time, the weaker examples shown in the rightmost column show that some topics are less interpretable and mainly consist of general legal terms, older function words, or OCR-related noise. All models managed to create a topic for German language, mainly using German stopwords. This is because only Swedish stopwords were considered in the preprocessing steps. We can also note that since lemmatization was not utilized in this thesis, some topics contain the same word multiple times in different forms, such as “skog” and “skogen”.

These examples are not used as an additional evaluation metric, but help illustrate what the automatic coherence and diversity scores represent in practice.

Table 4.11: Sample of topics and their top words from the different models. Table shows five interesting topics per model and one weaker topic listed furthest to the right. The displayed words are selected from each topic's highest-ranked topic words.

BERTopic					
krigs artillerie götha regements krigsrättens rättens finland	transport bord lod fat lock gamla	nedannämnde andel grufvebrytning grufvor masugn smedja	riksdaler tusende hundrade ränta banco riksgälds	und die mit undt worden wie	drake ukslag arken utalas stake
CTM					
hofrätten besvär ansökning wite ingifwen bewis	transport bord gamla diverse glas koppar	ting bergs påstående års tinget käranden	brott afstraffas slag förmågo straff dömd	mit worden und kommen alles die	män will huadh gudh haffwa ähre
ETM					
sahl öre interesse sal penningar betalt	transport gamla gammal bord lod fat	hemman skog bruk malm skogen måns	hustru tagit swarade wittnet wittnen berättade	und die mit worden undt will	anseende skal lag sådan böra smt
LDA					
banco riksdaler daler böta sedlar balken	bruk inspectoren bruks brukspa- tron bruket hammar	hemman skog hemmanet ägor skogen skatte	riddaren majoren lieutenan- ten general capitainen presidenten	und die mit worden undt wie	vore målet nämnde afseende anledning enär

4.3. Extendability to full dataset

This section explores how the models perform when applied to the full dataset, rather than only the 150k sample described in Section 3.1. The models presented here use the same configuration as the best performing models used in Section 4.2.1. However, the models were not all applied in the same way. BERTopic and CTM were fitted on a 150k sample and then used to assign topics to the full dataset, while LDA was trained directly on the full corpus. The results should therefore be interpreted mainly as a scalability and full-corpus assignment test, rather than as a direct replacement for the main sample-based evaluation.

Initially, there were problems with generating the embeddings for the full dataset in a reasonable time, therefore the main evaluation was performed on a smaller sample. At the end of the project, the embeddings were able to be generated, which allowed for this smaller section to explore the scalability of the models on a larger dataset.

Due to time constraints, the models are only run once in this section, instead of taking the mean from 3 runs, as done in Section 4.1. Thus, this section is only meant as an exploratory analysis and should not be considered central compared to the results in Section 4.1 and Section 4.2.

The OCTIS implementation of ETM was unable to complete the full dataset run because it failed during extraction of the full document-topic matrix. ETM is therefore omitted from this comparison. To not shift focus from comparisons between different models, BERTopic variations will not be considered in this section. Thus, this section will only cover the base BERTopic, CTM and LDA.

By comparing Figure 4.23 and Figure 4.19 we can see that BERTopic and CTM perform similarly when applied to the full dataset compared to the 150k sample. BERTopic scores slightly higher and CTM slightly lower. LDA however, shows a clear decrease when run on the larger dataset.

Figure 4.24 shows the diversity scores. CTM achieves a perfect diversity of 1.0. BERTopic is similar to its sample run in Figure 4.20 but LDA has increased.

In terms of runtime, Figure 4.25 shows that the full dataset experiments completed within a reasonable time. Compared to the runtimes in Section 4.2.1,

all runtimes increased, but LDA increased the most and became the slowest included model.

Note that the full dataset run values are not an average of multiple runs but only a single run. The results should thus not be analyzed in detail but only serve as a guideline for what changes running on the full dataset may incur.

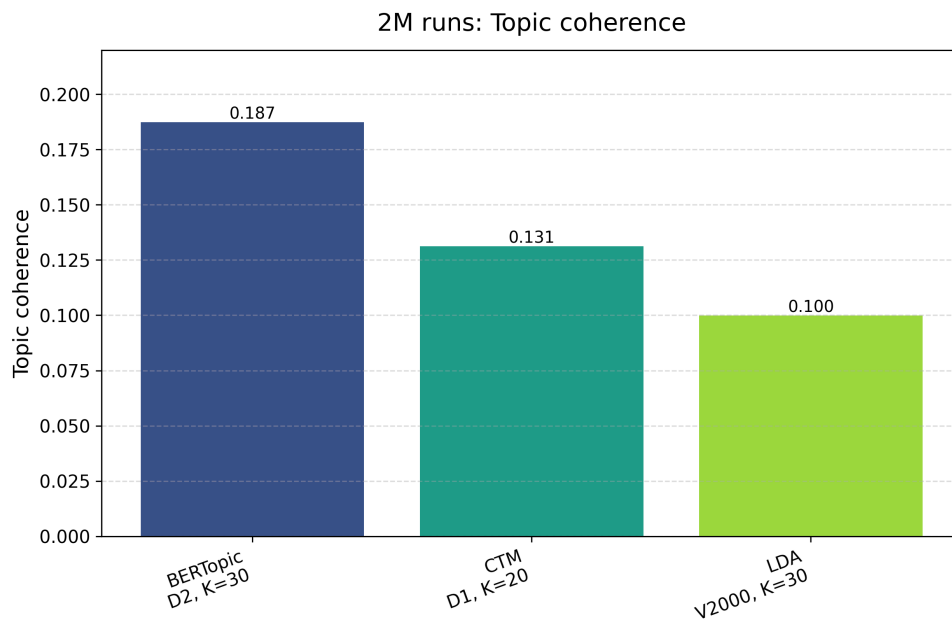


Figure 4.23: Models compared by coherence, when applied to the full dataset using the configurations that performed best on the sample run.

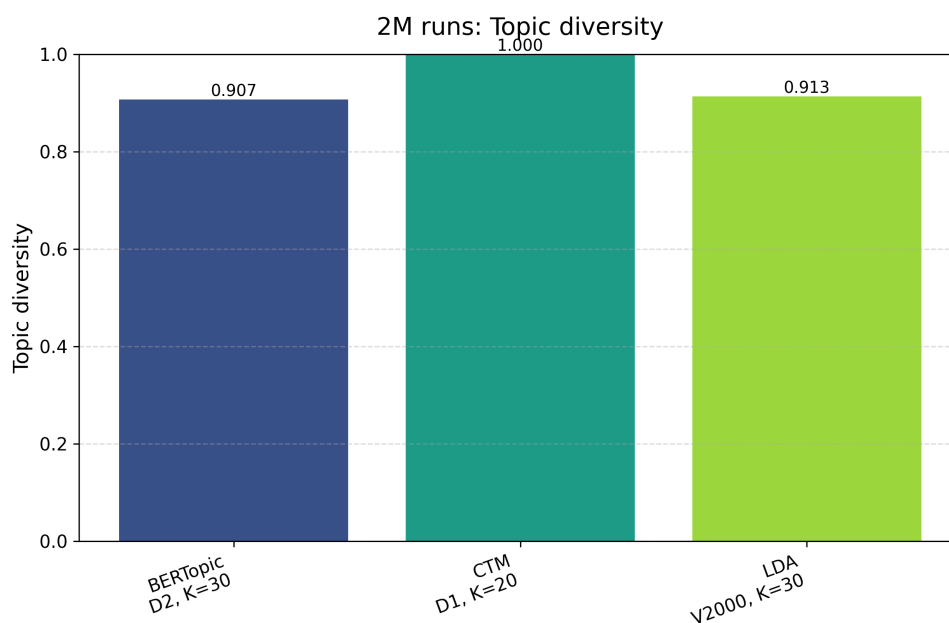


Figure 4.24: Models compared by diversity, when applied to the full dataset using the configurations that performed best on the sample run.

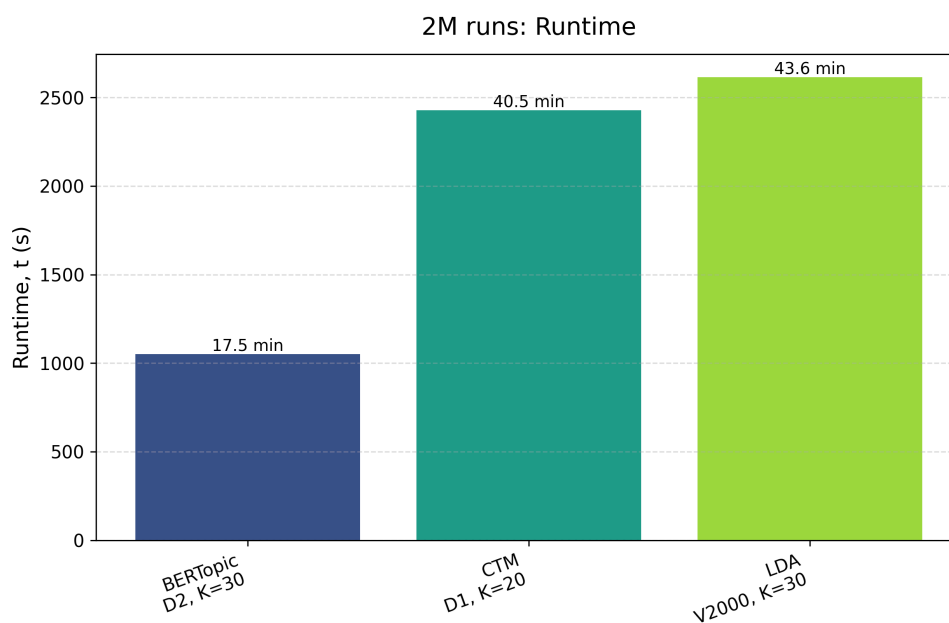


Figure 4.25: Models compared by runtime, when applied to the full dataset using the configurations that performed best on the sample run.

The amount of documents classified as noise by BERTopic can be viewed in Table 4.12 along with the noise by the corresponding BERTopic configuration on the dataset sample. The numbers for the sample run are taken from Table 4.2.

The table shows that the full dataset assignment produced a higher noise rate than the corresponding sample based BERTopic Base D2 $K = 30$ result, increasing from approximately 58% to 65%.

Table 4.12: The amount of documents classified as noise by the base BERTopic model when applied to the full dataset compared to the corresponding sample-based run.

Dataset	K	Total Docs	Noise Docs	Noise %
Full	30	1 774 919	1 154 652	65.05
Sample	30	150 000	86 591	57.7

Chapter 5

Discussion

This chapter discusses the results presented in Chapter 4. The discussion first returns to the research question and hypothesis stated in Section 1.2, before interpreting the main findings from the model comparison. The chapter then discusses the trade-off between coherence, diversity, and the role of noise handling in BERTopic. This is then followed by the influence of preprocessing, and then how many topics are appropriate for the models. The main analysis is based on the sample runs, but Section 5.7 covers how the results from the full dataset runs should be interpreted. Finally the implications and recommendations for Riksarkivet are discussed followed by a summary.

5.1. Research Question and Hypothesis

The research question presented in Section 1.2 of this thesis was: *What topic modeling techniques produce the best results on the dataset at Riksarkivet?* Based on the results in Section 4.2, BERTopic produced the strongest overall results. The best-performing configuration was the Base BERTopic model when using the D2 embeddings for $K = 30$.

This result partially supports the hypothesis that BERT-based models would outperform the alternatives. Both BERTopic and CTM performed competitively, with BERTopic achieving the highest coherence and CTM achieving the highest diversity. This reinforces the idea that contextual models are better suited for the noisy and historically varied documents used in this study.

The results also show that the advantage of neural or contextual models is not automatic. BERTopic performed well, but even though CTM achieved the next highest score, it was still a notable decrease compared to BERTopic and was more similar to the LDA. This suggests that even though BERT-based models are a good fit for the dataset, it is specifically BERTopic that stood out. Other methods should still be considered as options.

CTM also achieved high diversity but required substantially longer runtime. ETM, despite being a neural topic model, performed worse than LDA in terms of coherence.

The drawback of BERTopic is that almost half of the documents are classified as noise instead of being assigned to a topic. This ensures a higher level of relevance for the documents in the topics. If Riksarkivet, however, finds it important that all documents are classified into topics, using the reassignment strategy for BERTopic shows the second best overall results. Though it decreases diversity, it only scores slightly lower than the base model in coherence.

Based on these results, BERTopic is the most suited model for the dataset at Riksarkivet. Not only because of its high coherence, but because it manages to balance it with both a high diversity and the lowest runtime. It can also be easily modified to ensure no documents are classified as noise without a major score penalty.

5.2. Coherence Scores

BERTopic scored the highest in coherence. This is consistent with the findings of [52], where BERTopic also scored visibly higher compared to LDA. The results of this study still fall short compared to the best result of around 0.21 TC achieved by [52], but are still toward the higher end, outperforming every model except their very best. The coherence of the LDA also exceeded the values presented by [52], indicating that the dataset of Riksarkivet may be more well behaved than other noisy and historical datasets where topic modeling has been shown to work.

CTM managed to score higher than all models except BERTopic. This is also in line with previous studies. [6] presented results showing CTM scoring higher than both ETM and LDA, with ETM falling behind LDA. This study showed a similar result, but CTM and LDA were much more equal here compared to other studies.

The paper introducing BERTopic [9] also compared it to CTM. Their results showed, similar to those of this study, that BERTopic achieves a higher coherence than CTM. In their case, however, the difference was large, with BERTopic achieving almost twice the score of CTM. In this study the difference was notable, but CTM still achieved competitive scores in comparison.

The overall scores are not only comparable to the scores presented in [52], which also handles historical data, but are also competitive with the scores from [6] as well as the papers that introduced the models. This shows that, at least according to the evaluation metrics, this study is able to achieve results on the problematic dataset at Riksarkivet that are comparable to those achieved by other studies on more well-behaved datasets.

The LDA scores unexpectedly well, showing results competitive with the CTM. This was unexpected as it was expected to struggle with the noisy nature of the dataset as well as the large vocabulary, which it mostly has to disregard as it only considers a small portion of it.

The LDA's success might be explained by it benefiting from only considering a small subset of the vocabulary, thus disregarding a large amount of noise. This should not be the full explanation, as it scores much better than the ETM, which does the same thing. Interestingly, though, the ETM scores notably worse when attempting to utilize a larger vocabulary.

5.3. Diversity Score

The diversity score, measuring how distinct the topics were from each other, also showed similar results to previous studies.

BERTopic, which scored the highest in coherence, does fall behind CTM in diversity. This aligns with the results presented in [9] where The newly introduced BERTopic exceeded the CTM in coherence and while still achieving a competitive diversity, does not manage to surpass the CTM. Even in the study of [6], CTM scores the highest in diversity compared to both the ETM and LDA. ETM also performed particularly bad in diversity in their study which matches our results. Particularly, the ETM had the most visible K dependent diversity out of all the models in this study.

Though the ETM has been shown to perform bad in [6], the poor results in this study is still somewhat unexpected as the dataset description does seem to cater towards its strengths of being able to handle larger vocabularies and an abundance of rare words described in [2].

All models showed a drop off in diversity as the number of topics increased, except for the LDA. This made intuitive sense as it becomes harder to keep topics distinct and the room for redundancy increases as more topics are

created. ETM displayed the most clear example of this relationship as seen in Figure 4.14 where the diversity quickly drops.

5.4. Noise Classifications in BERTopic

A central finding is that BERTopic's treatment of outliers strongly affected its usefulness. The base BERTopic model classified a large proportion of documents as noise, in many cases close to or above half of the dataset. This is problematic because the purpose of topic modeling in this thesis is not only to create coherent topics, but also to organize documents in a way that could support future information retrieval.

The modified HDBSCAN parameters reduced the number of noise documents from around 50% to 35-40%. This is a notable decrease but still a large amount of noise. The downside is that it came at the cost of a decrease in coherence. This indicates that simply forcing the clustering algorithm to accept more documents into clusters can reduce topic quality.

In contrast, the reassignment strategy managed to remove the noise category while still preserving strong coherence. The results were however more varied across runs and different number of Topics. There was still a noticeable loss in diversity but altogether this makes reassignment the more promising BERTopic variation for this dataset.

Whether the different strategies for noise reduction should be used at all is up to the end user. If full coverage is of great importance then the reassignment strategy can achieve that without interfering with the quality too much. If quality of the topics are of priority then it is better to stick with the base model in order to maximize the coherence and diversity. For archival purposes, it might be preferable to ensure full coverage, but to allow for efficient retrieval for the end user, too much coverage may become counterproductive and decrease the usefulness of topics.

5.5. Heavy preprocessing vs Semantic Structure

This study also aimed to investigate the difference between different levels of preprocessing when generating the embeddings of BERTopic and CTM. The models were constructed with the aim of being able to utilize the semantic context of datasets instead of spending effort to clean up and remove stop-words in datasets. This comparison was inspired by the study by [52] where

they notices that BERTopic appeared to perform better when the data was more preprocessed.

The results of this study strengthen the claims that BERTopic may benefit from a more preprocessed dataset. Its coherence greatly benefited from using more heavily preprocessed data to generate its embeddings. This shows that it is able to extract semantic relationships even without the stopwords that are usually seen as important for binding text together, and does so more effectively.

This improvement might be explained by the fact that the OCR noise is so high that it disrupts the semantic flow to the point where removing the majority of the words actually leads to an improvement. This lets the models focus on the more context-heavy words instead of being affected by potential noise or spelling variations.

Interestingly however, when using the BERTopic version with modified HDBSCAN parameters, using the more preprocessed data actually decreased the coherence score of the model, while the version using the less preprocessed dataset was less affected. This may indicate that the stopwords play an important role when differentiating more noisy and vague documents.

When using the reassignment strategy for BERTopic, the effects are less clear, as the coherence score is more varied across number of topics and preprocessing level. No clear pattern emerges as the heavily preprocessed embeddings sometimes perform better, sometimes the same, and sometimes worse.

In the case of the CTM, no notable difference was observed between using D1 and D2 for the embeddings. There was no benefit but no drawback either. This suggests that the CTM is limited by other factors such as the noise in the dataset or the historical context, and the change in embeddings makes little difference in comparison.

The limited effect on CTM may also be explained by the fact that not enough heavy preprocessing was done. The stopwords removed were based on a single stopword list for modern Swedish that was combined with a list generated by a large language model based on the aforementioned list and words found in the corpus. The stopword removal in the study was

limited, so the fact that such a large difference was observed in BERTopic is noteworthy.

Overall it is surprising that removing the stopwords that play an important role in keeping the semantic structure of a text actually can benefit a model designed take advantage of them. This shows the contextual power that can be stored in embeddings and how they can extract meaning that does not always make sense by human merits but still produce powerful results.

5.6. Optimal Number of Topics

The optimal number of topics is not a conclusive result as multiple values for K all showed similarly high scores. Considering the variance in the scores of BERTopic across multiple runs, more runs would be required to confidently proclaim the best suited K .

BERTopic and CTM appeared to score slightly lower for $K = 10$, suggesting that 10 topics might be too small to properly explain the dataset, but there is no notable drop-off in coherence as topics increased. This may suggest that the dataset is rich in topics and may hint at value in evaluating the dataset at higher numbers of topics than just 60.

In terms of diversity, as described in section Section 5.3, most models did show a drop-off in diversity as the number of topics increased, which would suggest that a lower number of topics would be preferable. With the sole exception of the ETM, the drop-off is not severe and remains within acceptable bounds.

LDA shows almost no drop-off at all and the CTM stays rather stable, although there is a small decrease. BERTopic does show a non-negligible drop-off. It is not problematic in the evaluated range, but if the trend continues, it would probably not be useful to go past a certain number of topics.

5.7. Full Dataset Performance

The models were also evaluated on the full dataset as seen in Section 4.3. The full dataset runs were single runs, instead of averages of three as in the sample runs. This makes it difficult to draw clear conclusions from the results as they cannot be as reliably compared to the sample runs or even to each other. We can however observe the overall metric scores and draw

conclusions from if they appear similar or are vastly different from previous runs.

The results show that BERTopic and CTM were able to handle the full dataset without larger issues as both their coherence and diversity is similar to the sample runs. The diversity increased for both the models but since the change is not very big, it is hard to draw any definitive conclusions.

It is also noteworthy that CTM got a perfect 1 in diversity. This means that no word appears twice across any of the topics. This should however be unrelated to the size of the full dataset as both CTM and BERTopic still generate the topics based on a 150k sample.

The same partially goes for coherence, as it is like diversity, based on the topics generated from the sample. The co-occurrence of words is however measured across the entire dataset.

LDA and ETM attempts to generate the topics based on the whole dataset at once, instead of just a sample like CTM and BERTopic. This was needed as once the ETM and LDA models are fitted, they cant be extended with more data.

The LDA showed larger changes compared to its run on the sample. The coherence has decreased notably while the diversity increased. This follows the trend of increased diversity by the CTM and BERTopic, but it now comes at the cost of coherence. This means that the topics are very diverse but less meaningful.

ETM was unable to finish the full dataset run. The OCTIS implementation used was unable to fully extract the document-topic matrix. It may be possible to utilize the ETM by modifying the OCTIS implementation, but that was not explored in this thesis.

Lastly, The number of documents labeled as noise by BERTopic was shown to have increased. Compared to the sample run, the number has increased from ~58% to ~65.1%. This could be an indication that BERTopic doesn't fully generalize to the full dataset from just the 150k document sample. Thus, it may be beneficial to base the topics on a larger sample before applying the model to the rest of the dataset.

5.8. Practical Implications for Riksarkivet

For Riksarkivet, the results suggest that topic modeling is a promising approach for organizing OCR-transcribed historical Swedish documents. Using the standard BERTopic appears to be the most suitable model among those tested, because it combines the highest coherence with reasonable runtime and diversity.

Based on the metrics the results are competitive with studies performed on other datasets. This includes both other historical datasets as well as contemporary and well behaved datasets without OCR noise.

However, the results also suggest that topic modeling should be used carefully. The quality of the topics depends on preprocessing, noise handling, and the number of topics. In a practical archive system, the model should probably be combined with manual inspection or domain expert feedback before being implemented for the official retrieval or browsing functions. It may also be useful to run the model multiple times to maximize the results due to the stochastic nature of BERTopic.

When applying BERTopic to the full dataset, it must first be fitted on a smaller sample which is then used to assign topics to the rest of the dataset. To ensure the topics generalize to the entire dataset, the sample must be sufficiently large. The optimal size of the sample is not explored in this thesis and future work is therefore encouraged to try larger sample sizes.

Before applying the models, more advanced stopword removal should be applied. This study only performed stopword removal in a limited form. Putting more effort into it should not only produce higher results on the metrics, but also provide cleaner topics overall, as the current topics contain a large amount of noise and unimportant words.

The findings also indicate that models should not only be evaluated by coherence and diversity. Since the final goal is human use, future applications at Riksarkivet would benefit from human evaluation, especially by archivists or researchers familiar with the document collection.

5.9. Summary of the Discussion

In summary, the discussion shows that BERTopic gives the overall best performance when considering coherence and diversity together. This is

followed by CTM, LDA, and finally ETM. If diversity is of greater importance, then CTM is instead the strongest model.

If full topic coverage of the data is required, BERTopic can be modified to reassign all unassigned documents to the most fitting topic. This mainly comes at a cost of lowered diversity but still keeps the high coherence of the base model. If the loss in diversity is considered too high, the CTM is instead the better alternative to use.

The use of the more heavily preprocessed D2 embeddings showed clear improvements in both coherence and diversity for the base BERTopic model. These patterns were, however, not visible in any of the other models, with the CTM showing virtually no difference between the D1 and D2 embeddings.

Running the models on the full dataset showed that BERTopic and CTM can be applied to a larger dataset by fitting on a sample and assigning topics to the remaining documents. LDA was able to train directly on the full dataset and produced usable results, but did not show a clear advantage over BERTopic or CTM. ETM was less practical in the full-dataset setting because the OCTIS implementation failed during full document-topic matrix extraction.

The discussion shows that the evaluation results depend on the intended use of the topic model. The base BERTopic model produced the strongest intrinsic results, while the reassignment variant provided complete document coverage with only a moderate loss in diversity. CTM remained important because of its high diversity, and LDA showed that classical models can still be competitive on this dataset. These findings indicate that the best model is not determined by a single metric alone, but by the balance between topic quality, diversity, runtime, and document coverage.

Chapter 6

Conclusions and Future Work

6.1. Conclusions

In conclusion, BERTopic appears to be the most suitable model for the dataset at Riksarkivet. It provides the strongest coherence accompanied by high diversity. Additionally, it also runs the fastest, given that the embeddings have been generated.

BERTopic can also be modified to ensure even noisy documents are given a topic at minimal penalty to topic quality.

CTM falls behind BERTopic in terms of coherence but was shown to be the strongest model in terms of diversity instead.

LDA was shown to still produce competitive results on this dataset despite the struggles it was predicted to face. This showed that even though context based models performed the best, classical models should not be entirely disregarded.

Finally, this study showed that BERTopic, when using its base configuration, consistently performs better when using embeddings generated without stopwords, even when they are only partially removed. This is beneficial as it reduces load and allows the model to fit more relevant content words within its context window.

6.2. Limitations

The main limitation is that the evaluation is primarily based on intrinsic metrics. Topic coherence and topic diversity are useful for comparing models, but they do not fully capture whether the topics are meaningful to human users. A model may score well while still producing topics that are difficult to interpret, too broad, or not useful for information retrieval. There is also the added dimension that neural models sometimes score lower in the metrics because they might group topics based on wider context not measured by the metrics. In a larger study, a set of human evaluators could have been utilized to provide human input as a contrast to the intrinsic metrics.

Another limitation is that the results are based on a specific sample of the entire dataset. Generating the BERT embeddings took a substantial amount of time, so running it a large number of times or for the whole dataset proved problematic within the time limits of this study. Access to GPUs was also limited, which made longer runs difficult to complete. Different preprocessing choices, embedding models, hyperparameters, or document samples could change the ranking of the models. This is especially important for BERTopic when it comes to handling noise. This study only tested one specific configuration of modifying the HDBSCAN parameters, but it is possible that another configuration might produce very different results.

The dataset itself is challenging because it contains OCR errors, historical spelling variation, and Swedish linguistic properties such as compounds and inflections. These factors make it difficult to know whether lower scores are caused by model limitations, preprocessing limitations, or the inherent difficulty of the data.

This study was also done without expert knowledge of the dataset. A better understanding of the content of the dataset could have guided the evaluation more around details such as an expected optimal number of topics and expected content of said topics. The stopword list generated could also have been more curated with more knowledge of the dataset to remove dataset specific stopwords.

6.3. Future Work

This study compared multiple different models and how they performed on the specific dataset at Riksarkivet. This provided insight into what worked well and which models struggled. There are, however, other relevant models that were not compared in this study. One such model is Top2Vec. This model has historically been compared alongside BERTopic, as they are both able to capture context when creating topics. Previous work usually shows that BERTopic creates higher quality topics, but due to the uniqueness of this dataset, it would still be valuable to compare it alongside the other models in this study. A future study would therefore be encouraged to add Top2Vec to the models to be evaluated.

The main experiments in this thesis used a 150k sample of the dataset. Only one run was made for each model on the full dataset. This makes it difficult to compare the full dataset results to the smaller sample runs, or even directly

between models. In future work, it would be beneficial to focus more directly on the full dataset runs. For BERTopic and CTM, it would also be valuable to fit the models on larger samples before applying them to the full dataset. The results indicate that at least BERTopic may benefit from a larger or more representative fitting sample, since the full-dataset run assigned a larger proportion of documents to the noise category.

There are several improvements that could be added to this study by having expert knowledge of the data. As described in Section 6.2, it would be interesting to see the improvements when using a more carefully crafted stopword list. Using lemmatizers to preprocess the data could also be an interesting extension. It was disregarded in this study due to the complexity of applying it to both noisy and historical data.

Expert knowledge would also have opened up the possibility to explore seeded topic models. As described in Section 2.6.5, seeded models are a useful option when working with historical data as they allow the models to form topics around certain words and also bind together concepts represented by different words across different times.

Observing the percentage of documents labeled as noise for BERTopic-Base in Table 4.2 shows that when using the D2 embeddings, the noise is notably lower for $K = 10$ and $K = 50$. This is because one of the three runs used to calculate the average only classified below 10% of documents as noise. This seemed to occur randomly sometimes but has not been investigated further in this study. These runs appeared to be associated with lower overall metric scores but could still be interesting to investigate further in a future study.

6.4. Reflections

This section covers reflections of the project in different aspects.

6.4.1. Personal

My first thought about completing this project is that I am satisfied that it was able to produce sensible results that can be compared with previous work. Both the coherence and diversity scores were comparable and competitive with those of other studies.

This project also helped me value the importance of putting a lot of work into performing a pre-study on the material. Prior to this thesis, the topic model I had the most experience with was the ETM. The ETM later turned out

to be the worst performing model in the study. Discovering BERTopic was rather straightforward, as it is one of the most popular and recent models. The CTM, however, was not as visible and was a model I had not heard about before this thesis. CTM was only something I learned about when reading previous literature and comparative studies. Even though CTM did not perform as well as BERTopic, it still contributed a lot to the study by displaying the highest diversity score among all the models.

Additionally, one of the central comparisons performed in the thesis was that of using D1 or D2 embeddings. This was an aspect which I had not even considered until rather late in the research process, when I came across the study by Murugaraj et al. [52] while reading about topic models on historical data. Testing this turned out to be one of the more interesting parts of the study, as it was nonintuitive that BERTopic would actually perform better when removing semantic structure from the documents.

6.4.2. Technical

Considering the fact that both CTM and BERTopic outperformed LDA and ETM, and that ETM was less practical on the full dataset at Riksarkivet, it might have been more fruitful to focus on evaluating more of the modern models. I mentioned Top2Vec in the discussion and how that might have been worth trying on the dataset as well. It also uses embeddings, so it could have utilized the BERT embeddings. It could also have been interesting to try different language models for generating the embeddings, but using BERT is probably the most promising approach since there is KB-BERT trained on Swedish by Kungliga Biblioteket.

I was satisfied with the effectiveness of the stopword removal, even though it was not done with much domain expertise. As described in the discussion, I only used a simple stopword list for modern Swedish, and then used generative AI to make a corresponding list for older Swedish. I also added many words from the topics that were generated during trial runs. The list was not complete, as the final topics still contain many words that should be considered stopwords or removed for other reasons. The process of removing stopwords this way could probably go on for a very long time. This is why I am satisfied that the stopword removal was still helpful for the models.

It was difficult to run the data on the full datasets. When I first tried to generate the embeddings, it was estimated to take 48 hours for just the light

embeddings. Thus, I opted to keep all my tests to a 150k sample of the data. This worked fine for getting my results and comparing the models. When I tried again at the end of the project, it only took around 4-5 hours, which was a reasonable time. I do not know what changed to cause this difference, or if it had something to do with the GPU I was using.

Regardless, ETM was less practical in the full-dataset setting. The OCTIS implementation failed when extracting the full document-topic matrix. CTM and BERTopic were able to fit on a 150k sample and assign topics to the full dataset, while LDA was actually able to fit directly on the entire dataset. It is possible to apply ETM to the full data, but it has to be done one part at a time and comes with a lot of work. It could have been valuable to explore but would have been outside the scope of this thesis.

I have mentioned it earlier in the report, but the evaluation metrics used in this study are not perfect. As described in the background, they are sometimes less accurate on contextualized models compared to other models. They also create a gap between recorded values and actual human interpretation of the results. It would have been very interesting to attempt to apply human evaluation to the results. Though this would not have been impossible, it did not feel feasible for me to organize in the timeframe of this thesis. Most importantly, however, human evaluation is not perfect, as it introduces uncertainties in how the results should be interpreted and is overall more difficult to handle than the more straightforward intrinsic metrics.

6.4.3. Ethics and Sustainability:

The project does not cause any obvious concerns in terms of ethics. The mission of Riksarkivet is to preserve important documents for use by both institutions and individuals. It is a benefit to all of society and this project's objective which is to improve its search functions only makes it more efficient for its users. Making the search function easier to use may also allow more people to take advantage of the service.

In terms of sustainability, an increased efficiency in information retrieval will lower the computational load of the task. This may however cause the use of Riksarkivets services to increase which may in turn increase the total computational power needed. The societal usefulness of the service will be considered to outweigh this risk especially as the potential increase in computational power needed is not comparable to the amounts used in other industries.

The data used is public by default and thus does not expose any sensitive information.

6.5. Code Availability

The code and data presented in this study is available at the online repository [here](https://github.com/FOISALL/DA233X-Topic-Modeling-Riksarkivet) : <https://github.com/FOISALL/DA233X-Topic-Modeling-Riksarkivet>.

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models: LDA, Embedded Topic Model (ETM), Contextualized Topic Model (CTM) and BERTopic. The results show that models able to utilize semantic context perform better on the dataset at Riksarkivet compared to traditional frequency based models. BERTopic achieved the highest topic coherence, while CTM produced highly diverse topics but required substantially longer runtime. LDA remained competitive as a traditional baseline, while ETM performed worse on the evaluated metrics and was less practical in the full-dataset setting. The results also show that preprocessing remains important even for context-based models. In particular, BERTopic performed better when embeddings were generated from more heavily preprocessed data, despite the common assumption that retaining more semantic context through stopwords should benefit embedding-based methods.

"Keywords[eng]": "topic modeling, natural language processing, historical data, noisy data, optical character recognition, OCR, LDA, ETM, CTM, BERTopic, BERT",

"Abstract[swe]": "<p>Studien använder ett historiskt dataset från Riksarkivet, huvudsakligen på svenska, som innehåller en stor mängd brus från optisk teckenigenkänning, samt stora språkliga variationer till följd av språkförändring över tid.</p><p>Studien använder ett historiskt dataset från Riksarkivet, huvudsakligen på svenska, som innehåller en stor mängd brus från optisk teckenigenkänning (OCR), samt stora språkliga variationer till följd av språkförändring över tid.</p><p>Temamodellering är en väletablerad teknik inom språkteknologi. Sedan introduktionen av Latent Dirichlet Allocation (LDA) har metoden varit ett kraftfullt verktyg för att organisera stora textsamlingar i olika kategorier, eller ämnen, baserat på deras innehåll. Traditionellt har denna typ av modellering använt sig av ordfrekvenser i dokumenten. Detta tillvägagångssätt har visat sig ha svårigheter med att hantera texter med stora mängder brus eller har stora språkliga variationer. Nyare metoder använder istället kontextuella språkmodeller såsom BERT för att kunna använda den semantiska betydelsen i texter. Denna studie undersöker hur moderna metoder för temamodellering står sig jämfört med traditionella metoder. Studien använder ett historiskt dataset från Riksarkivet, huvudsakligen på svenska, som innehåller en stor mängd brus från optisk teckenigenkänning (OCR), samt stora språkliga variationer till följd av språkförändring över tid. Fyra olika temamodeller utvärderas: LDA, Embedded Topic Model (ETM), Contextualized Topic Model (CTM) och BERTopic. Resultaten visar att modeller som använder texternas semantiska kontext presterar bättre på Riksarkivets dataset än traditionella frekvensbaserade modeller. BERTopic uppnådde de mest sammanhängande ämnena, medan CTM producerade mycket varierade ämnen men krävde betydligt längre körtid. LDA förblev konkurrenskraftig men visade inga tydliga fördelar över de nyare modellerna, medan ETM presterade sämre enligt de utvärderade måtten och var mindre praktisk vid användning på hela datasetet. Resultaten visar även att förbehandling av data fortfarande är viktig även för kontextbaserade modeller. I synnerhet visade BERTopic förbättrade resultat när inbäddningarna genererades från mer kraftigt förbehandlad data, trots antagandet att mer bevarad semantisk kontext från t.ex. stoppord bör gynna kontextbaserade modeller.</p>",

"Keywords[swe]": "Temamodellering, Språkteknologi, Historisk data, Brusig

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data, OCR, optisk teckenigenkänning, LDA, ETM, CTM, BERTopic, BERT"  
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